

WHAR Arena: Benchmarking the State of the Art in Efficient Wearable Human Activity Recognition

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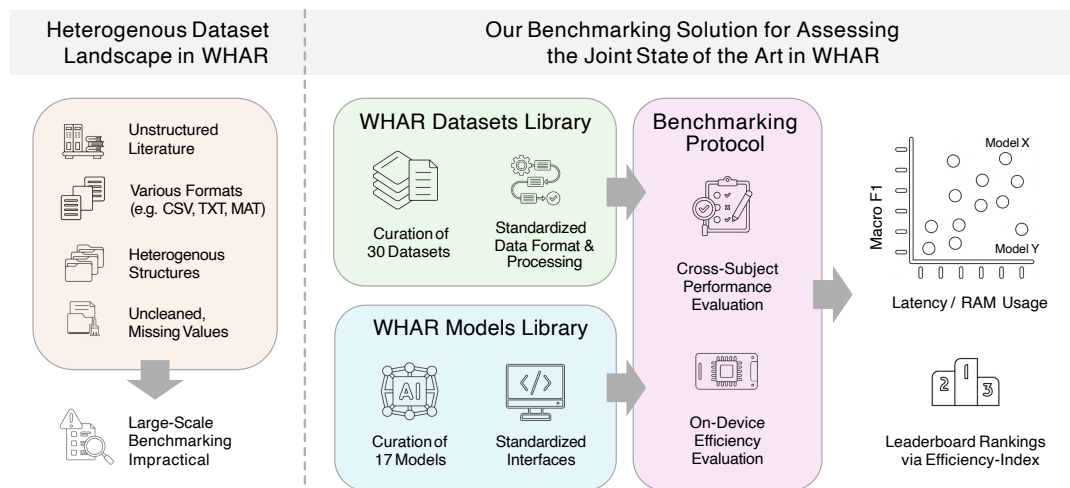


Fig. 1. Public Wearable Human Activity Recognition (WHAR) datasets are highly heterogeneous in terms of file formats, structure, and data quality, which makes standardized processing and fair comparison challenging. Our framework addresses this issue through two components: a WHAR Datasets library that parses and standardizes 30 datasets, and a WHAR Models library that unifies 17 representative models under a common inference interface. Together, these components enable large-scale benchmarking and reproducible analysis of performance–efficiency trade-offs in WHAR, culminating in robust relative leaderboard rankings.

Deep learning has become the dominant paradigm in Wearable Human Activity Recognition (WHAR), yet progress is obscured by a comparability crisis. Results are often reported using inconsistent datasets, custom data processing, and varying evaluation protocols, making state-of-the-art claims fragile. We address this with a large-scale, open-source benchmark that integrates 30 diverse datasets under standardized processing, unified model interfaces, and a shared cross-subject evaluation protocol. Evaluating 17 representative architectures across 4760 training runs, we jointly measure predictive performance alongside on-device latency, peak memory, and model size on an Android reference device. Our results reveal that the WHAR state of the art is distributed rather than dominated by a single architecture. While CNN-HAR achieves the highest mean macro-F1, top-performing models cluster tightly, indicating contemporary architectures have converged near a predictive performance ceiling. When accounting for deployment efficiency, compact neural models, such as TinierHAR, and classical Random Forests define the practically relevant Pareto frontier, whereas larger recurrent and hybrid models incur high hardware costs without corresponding performance gains. Consequently, while predictive performance has plateaued, substantial potential for future progress remains in optimizing deployment efficiency and improving adaptation to domain shifts. We release our full framework to support transparent reuse and extension.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Human-centered computing** → **Ubiquitous and mobile computing**.

Additional Key Words and Phrases: Human Activity Recognition, HAR, Benchmarking, Dataset Standardization, Data Processing, Wearables

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1 Introduction

Wearable Human Activity Recognition (WHAR) infers human activities from time-series signals collected by wearable sensors such as smartwatches [61], earables [44], and rings [65]. It is a core capability for applications in healthcare, fitness, sports, smart environments, and assistive systems [14, 43, 44, 81, 84], which operate in real-time and continuously. Current WHAR research is heavily influenced by the availability of open datasets [4], with classically used datasets such as WISDM [51] and OPPORTUNITY [85] complemented by newer datasets that cover a wider range of activity classes, sensor modalities, and configurations [20]. The need to support this increasing diversity and new compute capabilities of embedded devices has driven a shift from classical machine learning methods based on handcrafted features to deep learning approaches that model multivariate time-series data end-to-end using neural architectures incorporating convolutional, recurrent, and attention mechanisms [20]. While the field has shown increasing interest in efficient models for wearable and edge hardware, and continues to emphasize cross-subject generalization as a core requirement, comparability across studies remains limited by heterogeneous datasets, inconsistent data processing pipelines, insufficiently standardized cross-subject evaluation protocols, narrow dataset selections, and uneven reporting of on-device deployment-relevant efficiency metrics.

Despite this progress, predicting the actual performance WHAR under realistic conditions remains challenging. Unlike more mature benchmarking ecosystems in fields such as Natural Language Processing and Computer Vision [38], WHAR is increasingly affected by a comparability crisis [20, 26, 69, 109]. Commonly accepted benchmarks do not exist. The datasets used vary substantially in file formats, structural organization, and metadata, and in the absence of standardized data processing pipelines, studies typically rely on their own combinations of resampling, filtering, windowing, normalization, and augmentation [4]. As a result, these workflows tend to be highly customized and are frequently not documented in sufficient detail [69]. At the same time, evaluation protocols frequently fail to capture deployment-relevant generalization to unseen subjects. For these reasons, reported performance improvements may reflect differences in data preparation and splitting strategies as much as advances in model architecture. This issue is further compounded by comparisons that rely on a limited number of datasets, which fail to adequately capture the diversity of WHAR settings in terms of sensor modalities and configurations, number of channels, and activity classes, and which often use outdated baselines. Finally, existing benchmarks tend to focus primarily on predictive performance, although deployment efficiency is a critical requirement for wearable and edge applications [34, 117]. Consequently, state-of-the-art performance in WHAR should be defined jointly across these dimensions. Benchmarks that fail to do so, provide only limited evidence of meaningful progress.

This paper is guided by the research question: *What are the practical trade-offs between predictive performance and deployment efficiency for representative WHAR model architectures when evaluated under a unified, reproducible, and cross-subject benchmark across diverse WHAR settings?* To answer this question, we propose an open-source benchmarking framework that standardizes 30 WHAR datasets from a variety of settings, evaluates 17 representative WHAR model architectures under standardized data processing and cross-subject evaluation protocols, and jointly measures predictive performance and deployment efficiency, including latency and RAM usage. Our benchmarking framework is designed as a reusable basis for evaluating future methods under consistent experimental conditions.

Our core contributions are as follows:

(1) *WHAR Datasets Library*: We introduce an open-source library¹ that integrates 30 diverse WHAR datasets and makes them readily usable for benchmarking by converting heterogeneous raw data into a standardized format via dataset-specific parsers and configuration-driven processing pipelines. This design enables reusable and extensible workflows across studies, improves reproducibility, and helps address the comparability crisis. To support large-scale experimentation, the library provides efficient caching and framework-agnostic data loading, with adapters for PyTorch [77] and TensorFlow [1].

(2) *WHAR Models Library*: We introduce an open-source library² that assembles 17 representative WHAR model architectures with standardized interfaces. These models provide consistent baseline references for future studies, enabling systematic and comparable evaluations.

(3) *Benchmark Evaluation and Analysis*: We conduct a large-scale empirical evaluation comprising 4760 completed model-split training runs under a rigorous cross-subject evaluation protocol. Our evaluation jointly considers predictive performance and deployment efficiency. Based on these results, we derive Pareto fronts and construct robust leaderboards using a purpose-defined Efficiency Index to better characterize the state of the art in WHAR. To provide easier access to these findings, we

¹<https://github.com/teco-kit/whar-datasets>

²<https://github.com/teco-kit/whar-models>

host an accompanying website ³ featuring interactive data views, which will also serve as a living platform to publish future updates and leaderboard submissions. Ultimately, this analysis reveals that contemporary architectures converge toward a similar predictive ceiling.

2 Background & Related Work

To contextualize the need for our benchmark, this section outlines the current methodological landscape and critical shortcomings in WHAR. First, [Section 2.1](#) establishes the theoretical prerequisites for rigorous machine learning benchmarking. Building on these principles, [Section 2.2](#) diagnoses the comparability crisis currently present in WHAR. Finally, [Section 2.3](#) critically examines prior empirical benchmarking efforts, highlighting their limitations in scale, evaluation protocols, and joint assessment of predictive performance and on-device deployment efficiency. Together, this background demonstrates the need for the standardized, large-scale, and hardware-aware benchmarking solution that our work provides.

2.1 Theoretical Benchmarking Foundations

Benchmarks are empirical evaluation protocols that drive scientific progress by making results comparable across methods [38]. In machine learning, the foundation of this comparability is the holdout method, in which data are split into predefined, disjoint sets and final evaluation metrics are computed on a fixed, shared test set [38]. Within this framework, researchers are free to innovate in model architectures, learning strategies, and data preparation, the latter being a valid and often influential factor in performance, particularly in WHAR, where choices such as resampling, filtering, windowing, and splitting strongly affect the structure of the input data [42, 46, 69]. For this reason, when the goal is to compare model architectures in isolation, they must be evaluated under the same data processing pipeline, since variations in these steps can yield substantially different effective datasets and obscure whether performance improvements stem from novel model architectures or from data preparation.

Beyond fixed data splits, a valid benchmark must account for external validity [38]. Traditional statistical benchmarking often assumes that training and test data are drawn from the same stationary distribution, yet in practice deployed models inevitably encounter domain shifts. Therefore, properly assessing robustness requires benchmarks to explicitly simulate such shifts within their evaluation protocols [38]. In WHAR, domain shift is the norm rather than the exception, largely due to inter-subject variability in human physiology, movement patterns, and device placement [26, 69]. Consequently, to align with sound benchmarking principles, WHAR evaluations should prioritize generalization to unseen individuals by enforcing strict subject-level separation between training and test sets, ensuring that reported performance more faithfully reflects a model’s external validity under realistic deployment conditions.

WHAR encompasses a wide range of settings characterized by differing sensor modalities and configurations, varying numbers of sensor channels, and heterogeneous sets of activity classes [20]. This diversity implies that benchmarks should span a large and varied collection of datasets in order to adequately reflect real-world deployment conditions. In such heterogeneous environments, it is important to distinguish between absolute performance metrics and relative model rankings [38]. Absolute performance is often highly sensitive to domain shifts and dataset-specific noise, which are particularly pronounced across diverse WHAR settings [56], whereas evidence from more mature fields such as Computer Vision and Natural Language Processing suggests that relative model rankings are more stable across datasets [38]. Therefore, evaluating models across a multitude of datasets enables leaderboards that reflect consistent relative performance and provides more reliable guidance for practitioners by identifying architectures that generalize well across the broad WHAR landscape.

Consequently, a rigorous benchmark should integrate four core components: a diverse collection of curated datasets spanning varying target deployment environments (R1), a standardized data processing pipeline with fixed splits (R2), an evaluation protocol that accounts for external validity (R3), and a centralized leaderboard for objective comparison (R4). By standardizing the data preparation and evaluation process, the benchmark ensures that the object of comparison, such as a model architecture or a training protocol, is assessed under equivalent conditions. In WHAR, this entails enforcing consistent data processing due to its strong influence on input representations, ensuring strict subject-level separation to capture inter-subject variability, and spanning numerous datasets to represent the field’s varied sensor modalities, configurations, and activity classes. Without these shared components, meaningful comparison is not feasible and robust relative model rankings cannot be established methodologically.

³<https://teco-kit.github.io/whar-arena/>

2.2 The Comparability Crisis in WHAR

Despite the clear theoretical requirements for valid benchmarking outlined in Section 2.1, current WHAR practices routinely violate these principles, leading to a profound comparability crisis. The foundation of this crisis begins with the raw data itself, as WHAR datasets are characterized by extreme heterogeneity in file formats, directory structures, and metadata conventions [4]. A comprehensive survey by [4] revealed that nearly 68% of researchers require "some" or "a lot" of data processing effort just to make open datasets usable, with the lack of standardized formats and missing metadata cited as primary barriers. Because integrating new datasets requires substantial manual engineering, researchers are disincentivized from evaluating across multiple datasets covering a variety of WHAR settings (R1). Furthermore, the inherent subjectivity of human annotation protocols introduces systemic discrepancies in ground-truth labels, thereby substantially complicating the development of robust models from multiple data sources [69].

Beyond structural data formatting, the lack of standardized experimental data preparation severely undermines reproducibility [42, 69]. Despite the rapid advancement of model architectures in WHAR, experimental data preparation is often described only in limited detail. Because there is no universal approach, every research group implements its own custom processing pipeline. Disparate choices in resampling rates, noise filtering, and window segmentation strategies fundamentally alter the underlying signal distributions. This allows researchers to inadvertently tune data preparation to favor their specific model architectures. Furthermore, naive windowing strategies can cause data leakage between training and testing sets, artificially inflating performance metrics [46]. This comparability crisis is further compounded by the historical reliance of older datasets on precomputed handcrafted features, which are fundamentally incompatible with newer deep learning approaches based on end-to-end neural networks designed to process raw time-series data [109]. Ultimately, this systemic lack of consistency breaks the requirement for standardized data processing with fixed splits (R2). Because researchers do not evaluate their model architectures on identically processed data, absolute results cannot be compared across papers, even when evaluating the same dataset, obscuring genuine state-of-the-art progress.

Historically, evaluations in WHAR have also struggled to assess true external validity. Many studies have used random data splitting techniques that inadvertently mix samples from the same subject into both the training and testing sets [46, 112]. This practice explicitly breaks the requirement to simulate realistic domain shifts and measure external validity (R3). While the community has increasingly recognized this flaw in recent years and shifted toward stricter leave-one-subject-out cross-validation, a large portion of the existing literature still reports highly optimistic performance metrics that fail to reflect how models generalize to unseen individuals in real-world deployment.

Recognizing the broader comparability crisis, several works have proposed conceptual and methodological solutions. To support standardized dataset creation, [70] introduced a comprehensive framework and taxonomy and recommended the publication of data processing scripts, while [4] outlined an open dataset lifecycle to address heterogeneous formats. Other efforts have focused on practical standardization: for instance, [42] released a package that harmonizes hyperparameter settings, though it does not address data processing, and [46] and [104] manually standardized specific groups of datasets to mitigate evaluation biases. While these contributions diagnose the problem and provide useful methodological guidance, they do not provide a universally reproducible solution that can automate data preparation across the field.

Because the primary literature systematically violates standard benchmarking principles, secondary literature attempting to aggregate these findings is inherently limited. The extensive body of survey and review papers in WHAR (e.g., [20, 27, 30, 36, 47, 50, 80, 81, 88, 89, 99, 115]) provides valuable qualitative overviews and aggregates extensive dataset lists. However, their quantitative comparisons simply compile reported results from disparate papers. Since these numbers are generated using unstandardized splits, custom data preparation, and differing evaluation protocols, attempting to aggregate them into a trustworthy relative ranking is mathematically and methodologically unsound. In the absence of a unified benchmark, compiling such disparate results yields no comparative value, directly highlighting the critical need for a centralized, objective leaderboard (R4).

2.3 Existing Benchmarking Efforts in WHAR

While several empirical benchmarking efforts applicable to WHAR exist, they consistently fall short of the theoretical requirements for validity outlined in Section 2.1 (R1–R4). Furthermore, as highlighted in the introduction and detailed in Table 1, they routinely ignore the practical realities of edge deployment by failing to treat hardware efficiency as a primary evaluation target. Existing efforts can be broadly categorized into general time-series benchmarks, WHAR-specific benchmarks, and works targeting specialized sub-problems or edge deployments.

Recent large-scale benchmarks designed for general time-series classification, such as [29], operate at an impressive scale (29 datasets) and explicitly address structural challenges such as class imbalance. However, as shown in Table 1, only 7 of these datasets are specifically focused on WHAR. More critically, they lack WHAR-specific standardized data processing pipelines (R2), and the evaluated models are typically generic architectures rather than those optimized for the nuances of wearable sensor data. Furthermore, computational efficiency in these broader benchmarks is evaluated solely via parameter count, which fails to capture actual on-device deployment constraints.

Conversely, benchmarks tailored specifically to standard WHAR suffer from a severe lack of scale and therefore fail to represent the vast array of sensing modalities in the field (R1). As Table 1 illustrates, standard evaluations frequently rely on just 2 to 6 datasets [2, 40, 74, 75, 112]. Beyond scale, these works also violate fundamental principles of external validity. For example, [2] and [40] evaluate models without enforcing subject-level separation across splits, explicitly breaking the requirement to simulate realistic domain shifts (R3). Furthermore, Table 1 highlights a systemic flaw across the entire field: not a single existing benchmark provides a standardized, reproducible data processing pipeline (R2), making cross-study comparability impossible.

A third category focuses on highly specific, isolated challenges. This includes frameworks evaluating data ambiguity [32], domain generalization and adaptation [56, 68], or continual learning [45]. While methodologically valuable within their specific niches and generally better at enforcing subject-level separation, their restricted dataset selections (typically 6 or 7 datasets) limit their utility as state-of-the-art evaluations for general-purpose WHAR. They do not provide the centralized, comprehensive leaderboard (R4) required by practitioners selecting standard models.

Another distinct approach in the related literature involves long-term, community-driven competitive benchmarking. Initiatives such as HASC [48] and EVAAL [22] constitute longitudinal efforts that combine shared corpora with recurring international competitions. Unlike static dataset releases, these initiatives define repeated evaluation campaigns with standardized tasks, metrics, and evaluation environments. By evaluating submissions on unseen, held-out data, these efforts provide a competitive benchmarking framework that fosters a community ecosystem and enables the longitudinal comparison of algorithmic progress. While this paradigm successfully standardizes experimental setups for specific sub-tasks or ambient-assisted living scenarios, these competitions often remain constrained to their specific environments. Consequently, they do not offer the highly diverse dataset suite (R1) needed for generalized WHAR, nor do they typically address the joint evaluation of predictive performance and dynamic on-device efficiency required for modern edge deployment.

As detailed in Table 1, offline efficiency metrics (e.g., parameters, MACs, FLOPs) are reported inconsistently across existing benchmarking efforts. While some works explicitly target Edge WHAR [11, 116, 117], they largely overlook a systematic evaluation that jointly considers predictive performance and deployment efficiency. A few studies take an important step by conducting on-device efficiency experiments [56, 116], but even these lack a comprehensive analysis along this dimension. As a result, existing leaderboards remain fundamentally incomplete for real-world wearable applications, where models must simultaneously achieve high predictive performance and meet strict hardware constraints.

In contrast to generic time-series frameworks, narrow WHAR benchmarks, and incomplete edge evaluations, our framework directly addresses all four theoretical benchmarking requirements while closing the efficiency trade-off blind spot. As summarized in the final row of Table 1, we execute experiments across a large and diverse suite of 30 datasets (R1). To our knowledge, our framework is the first WHAR-specific benchmark to combine a strictly standardized, open-source data processing pipeline (R2), realistic subject-level separation for external validity (R3), and a centralized basis for trustworthy cross-dataset comparison (R4) at this scale. Crucially, we define the WHAR state of the art through the joint consideration of predictive performance and deployment efficiency. By providing standardized offline metrics alongside dynamic on-device measurements (inference latency and RAM usage), and by evaluating these trade-offs via a Logarithmic Efficiency Index and Pareto frontiers, our benchmark establishes robust rankings that reflect genuine readiness for real-world wearable deployment.

3 Methodology for Models

To construct the model suite used in our benchmark comparison, we followed a structured, literature-driven selection process focused on representative WHAR models. The goal was to cover a diverse range of approaches, including both classical methods and more recent architectures, in order to capture different modeling paradigms and design choices. Given the heterogeneity in how models are presented in the literature, it is challenging to identify all relevant approaches directly through keyword-based queries alone. Models are often introduced as part of broader systems, used as baselines, or embedded within application-driven studies, making them difficult to retrieve systematically in isolation.

Table 1. Comparison of existing WHAR benchmarking efforts. Prior works are limited in scale, lack standardized data preparation, and report efficiency metrics inconsistently with no systematic analysis of efficiency–accuracy trade-offs. In contrast, our framework operates at an unprecedented scale (30 datasets) and is the only benchmark to enforce standardized, open-source data processing pipelines. Furthermore, we provide comprehensive offline (Params, MACs, FLOPs) and dynamic on-device (latency, RAM) metrics to explicitly evaluate the trade-off between predictive performance and hardware efficiency for real-world edge deployment.

Reference	Focus	WHAR	Rigour	Baselines		Reproducibility		Efficiency Evaluation		
		Datasets	Subj. Sep.	Classical	Deep	Open-Source	Std. Processing	Offline	On-Device	Trade-off
Dempster [29]	Time-Series	7	✓	✓	✓	✓	×	✓	×	×
Abdel-Salam [2]	Generic WHAR	6	×	✓	✓	×	×	×	×	×
Hossain [40]	Generic WHAR	5	×	✓	✓	×	×	×	×	×
Lu [56]	Domain Gen. in WHAR	6	✓	×	✓	✓	×	×	✓	×
TinierHAR [11]	Edge WHAR	14	✓	×	✓	✓	×	✓	×	×
TinyHAR [117]	Edge WHAR	6	✓	×	✓	✓	×	✓	×	×
Napoli [68]	Domain Adapt. in WHAR	6	✓	✓	✓	✓	×	×	×	×
Geissler [32]	Data Amb. in WHAR	6	✓	×	✓	✓	×	×	×	×
MLP-HAR [116]	Edge WHAR	6	✓	×	✓	×	×	✓	✓	×
MLP-Mixer [74]	Generic WHAR	3	×	×	×	✓	×	×	×	×
DeepConvLSTM [75]	Generic WHAR	2	✓	✓	✓	✓	×	×	×	×
CNN-HAR [112]	Generic WHAR	2	×	✓	✓	✓	×	×	×	×
Ours	Edge WHAR	30	✓	✓	✓	✓	✓	✓	✓	✓

Therefore, we applied two complementary search strategies to identify suitable models. First, we conducted a systematic literature search following the PRISMA methodology, as described below, to obtain a comprehensive set of candidate model papers. Second, we performed backward citation tracking (backchaining) to identify additional relevant and influential architectures that may not have been captured through the initial search.

3.1 PRISMA Paper Retrieval

We applied an initial keyword-based search strategy on the ACM Digital Library (ACM-DL), which is a major repository for computing research and includes the vast majority of WHAR model papers. Based on influential papers in the field and preliminary scoping searches, we designed a search query targeting papers that mention Human Activity Recognition (HAR) in the abstract together with terms indicative of modeling contributions (e.g., model, framework, method, architecture, system, network). Concretely, we use the search query listed in Figure 2. This search returned 864 candidate papers. As this set contained many publications not relevant to wearable sensor-based HAR (e.g., video-based activity recognition or application-focused studies), we performed an additional filtering step in which we decided, based on the abstract, whether to retain or discard each paper. This left us with 83 papers for further inspection.

Abstract:(human activity recognition OR HAR)

AND

Abstract:(model OR framework OR method OR approach OR architecture OR system OR network)

Fig. 2. ACM Digital Library search query used to identify WHAR model papers.

3.2 Backward Citation Tracking

To enrich the collected list of candidate models, we performed backward citation tracking starting from the papers selected through the PRISMA search. This strategy allowed us to identify foundational and influential architectures from other sources that may not have been captured through the initial keyword search.

3.3 Model Selection

The search and backward citation procedures described above yielded a broad and heterogeneous set of WHAR model papers. To turn this set into a coherent model suite, we applied selection criteria that emphasize reproducibility, architectural generality, and practical relevance for a WHAR benchmark:

- (1) **Availability** We only include WHAR models in the final collection whose source code is publicly available. This helps ensure that the models are implemented as their authors intended, which in turn supports fair and reproducible comparison. In many cases, it would also not be possible to reimplement an architecture from the paper alone, as key implementation details are frequently missing.
- (2) **Task Fit** We restrict our selection to models that directly address supervised, multi-class classification of windowed time-series from wearable sensors. Architectures designed specifically for forecasting, anomaly detection, or multi-task

objectives without an explicit activity classification head are excluded, since they cannot be integrated into a unified WHAR evaluation protocol without substantial redesign.

- (3) **Architectural Generality** We exclude models whose architectures encode hard-wired assumptions about a particular dataset, such as fixed sensor-placement graphs, dataset-specific hand-crafted features, or bespoke activity taxonomies. Instead, we require architectures to operate on generic multivariate windows with a configurable number of channels, time steps, and classes, so that they can be instantiated across all datasets in our benchmark without structural changes.
- (4) **Computational Feasibility** Since our benchmark is designed with wearable and edge deployment in mind, we exclude architectures whose parameter counts or reported training requirements are clearly incompatible with WHAR-scale experiments (e.g., multi-hundred-million-parameter Transformers originally designed for vision or language). We require that models can be trained on all datasets within reasonable GPU memory and runtime budgets and that they expose a well-defined trade-off between accuracy and resource usage.

The resulting set of candidate architectures was then reviewed manually and consolidated into a representative model suite. In total, we implemented a broader set of 19 deep architectures in our codebase, but selected only 14 deep models for inclusion in the final benchmark. Implemented models that imposed strong sensor-structure, task-structure, or preprocessing assumptions were excluded prior to benchmarking because they could not be evaluated fairly across the full dataset suite under one shared protocol. For example, DeepSense, GlobalFusion, IF-ConvTransformer, and AttenSense rely on architectural assumptions about sensor grouping or fixed preprocessing structures that are difficult to reconcile with a general WHAR benchmark spanning highly heterogeneous datasets. In addition, AROMA-Joint could not be adapted to a general WHAR benchmark setting without departing too far from the authors’ intended design. Finally, we included three widely used classical baselines (k-NN, Random Forest, and SVM) as reference points within the benchmark, resulting in a final suite of 17 benchmarked models. Table 2 summarizes the selected models along with the key metadata used in our evaluation.

3.4 Implementation Details

All deep models were integrated into a unified PyTorch [77] wrapper interface, enabling training and evaluation through a single pipeline and ensuring consistent and comparable experimental conditions. For models already compatible with the benchmark tensor shape, we kept the original architecture unchanged and added only wrapper-level input reshaping. For adapted models, we preserved the upstream model logic as closely as possible while relaxing hard-coded assumptions from the original repositories. To support reproducibility and reuse, we publish the source code with the integrated model implementations as the WHAR Models Library ⁴. We applied the following changes to the original implementations to make them compatible with our benchmark input format and training pipeline:

- **CNNHAR** [112] was ported from the original fixed-window implementation to PyTorch and modified with adaptive temporal pooling so that it could process variable sequence lengths.
- **Attend-and-Discriminate** [3] was run with a single benchmark-wide dropout configuration instead of the original dataset-specific tuning.
- **Guan-LSTM** [37] is included in the benchmark for coverage of this architectural line, but under our standardized training pipeline it is trained as a regular single model rather than as an ensemble.

4 Methodology for Datasets

The construction of the dataset suite for our multi-dataset benchmark is guided by three primary objectives: (i) inclusion of widely used, canonical datasets that have shaped the WHAR literature, (ii) incorporation of more recent datasets that reflect emerging sensor modalities and application domains, and (iii) coverage of a diverse range of WHAR settings.

To this end, our dataset suite includes established benchmarks such as WISDM [51] and UCI-HAR [6], which have been used extensively for model evaluation, as well as more recent contributions such as WEAR [14], which introduce novel settings and challenges. Beyond temporal diversity, our goal is explicitly to capture heterogeneity in application domains, including activities of daily living (ADL), locomotion, fitness tracking, sports analysis, fall detection, and health-related monitoring.

The dataset suite construction process consists of two main stages. First, we perform a broad and systematic dataset search to identify a comprehensive pool of candidate datasets (Section 4.1). Second, we apply a structured selection procedure

⁴<https://github.com/teco-kit/whar-models>

Table 2. Model overview with trainable parameter counts and approximate forward-pass FLOPs for C=6 sensors, K=5 classes, and T=128 timesteps. Attn = attention, CNN = convolutional neural network, Dense = fully connected layers, RNN = recurrent neural network, Graph = graph-based interaction, Spec = spectral preprocessing, and FE = feature engineering. Transformer-style attention is counted under Attn.

Model	Params	FLOPs	Attn	CNN	Dense	RNN	Graph	Spec	FE
Deep Models									
Attend+Discriminate [3]	371.5k	171.5M	✓	✓	✓	✓	×	×	×
CNN-HAR [112]	63.2k	11.3M	×	✓	×	×	×	×	×
DANA [58]	457.9k	102.4M	×	✓	✓	✓	×	×	×
DeepConvLSTM [75]	457.9k	176.2M	×	✓	✓	✓	×	×	×
DeepConvLSTM-Attn [66]	474.6k	179.9M	✓	✓	✓	✓	×	×	×
DeepConvShallowLSTM [13]	399.9k	207.6M	×	✓	✓	✓	×	×	×
DynamicWHAR [62]	1.12M	14.7M	×	✓	✓	×	✓	×	×
Guan-LSTM [37]	798.0k	205.6M	×	×	✓	✓	×	×	×
MLP-HAR [116]	157.7k	2.3M	×	×	✓	×	×	✓	×
MLP-Mixer [74]	605.9k	87.5M	×	×	✓	×	×	×	×
SA-HAR [57]	401.3k	121.5M	✓	✓	✓	×	×	×	×
TinierHAR [11]	7.3k	892.3k	✓	✓	✓	✓	×	×	×
TinyHAR [117]	159.1k	9.3M	✓	✓	✓	✓	×	×	×
Triple-Cross-Attn [103]	278.3k	17.9M	✓	✓	✓	×	×	×	×
Simple Models									
k-NN [78]	–	–	×	×	×	×	×	×	✓
Random Forest [78]	–	–	×	×	×	×	×	×	✓
SVM [78]	–	–	×	×	×	×	×	×	✓

(Section 4.2) to filter this pool according to criteria aligned with the requirements of our benchmark. This process results in a final dataset suite (Section 4.3) comprising 30 carefully curated WHAR datasets.

4.1 Dataset Search

Identifying relevant datasets in WHAR is inherently challenging due to the absence of standardized reporting practices. Unlike methodological contributions, datasets are not introduced or indexed consistently in the literature: some publications focus explicitly on dataset creation, while others introduce datasets only as a secondary contribution. Furthermore, commonly used keywords such as “dataset,” “benchmark,” or “database” are pervasive across WHAR publications, making it difficult to construct precise search queries that reliably isolate dataset contributions. As a result, applying a strictly formalized search protocol (e.g., PRISMA-style systematic reviews) is not feasible in this context.

To address these challenges, we adopt a two-stage search strategy. First, we collect datasets used in the model papers selected for our benchmark (see Section 3.3). As these works were chosen to represent influential and diverse approaches in WHAR, the datasets they evaluate reflect widely adopted and practically relevant choices in the literature. This step yields an initial pool of 23 datasets. Second, we extend this set by extracting datasets referenced in survey and benchmark papers identified during our literature analysis (see Section 2.3). These sources help identify datasets that are less frequently used in model evaluations but remain relevant to the domain. This approach ensures coverage of both historically influential datasets and newer or less frequently used ones, providing a comprehensive foundation for subsequent filtering and curation. By combining these complementary strategies, we assembled an initial pool of 81 datasets.

4.2 Dataset Selection

Given the large corpus of 81 datasets identified during the search phase, we define a set of filtering criteria to ensure the quality, relevance, and usability of the final dataset suite for our benchmark. The criteria are as follows:

Modality. We restrict our selection to datasets that contain time-series sensor signals related to human motion or physiology, as required for WHAR. This includes modalities such as accelerometer, gyroscope, and magnetometer signals (i.e., inertial measurement units, IMUs), as well as physiological signals such as electrocardiogram (ECG) and photoplethysmography (PPG). Datasets focusing exclusively on other HAR paradigms, such as vision-based, skeleton-based, video-based, or audio-based HAR, are excluded. For multimodal datasets, only the relevant sensor modalities are considered. Datasets without suitable modalities are filtered out (e.g., Epic-Kitchens-100 [25], Wetlab [87], and BON [102]).

Subjects. Since our benchmark focuses on cross-subject generalization, datasets must provide subject identifiers to enable cross-subject evaluation in realistic deployment settings. Datasets lacking subject-level annotations are excluded, as they do not support this form of evaluation (e.g., CHAD [91], PARDUSS [92], SisFall [100], and CSL-SHARE [53]). In addition, datasets must contain data from multiple subjects. Datasets with only a single subject (e.g., SKODA [97]) are also excluded, as cross-subject evaluation is not possible.

Availability. To enable a high degree of automation and reproducibility, we require datasets to be publicly accessible without restrictive access procedures. We therefore exclude datasets for which no download link could be found online, which require manual approval, or that involve registration barriers, since such restrictions hinder scalable dataset integration and future leaderboard submissions. Based on this criterion, datasets such as DA [93], PAR [73], CHARM [108], BijalHAR [12], MEAD [95], Stanford-ECM [67], and HOSPITAL [113] were excluded because access is not readily available online or requires manual requests. The same criterion also excludes datasets that require online registration prior to access, including DIP [41], HASC2010 [48], MobiAct [106], MobiFall [107], FallAllID [86], TNDA-HAR [52], MMAAct [49], and UESTC-MMEA-CL [111].

Size. We exclude datasets that are excessively large, as their inclusion would disproportionately increase training time, thereby limiting the number of datasets that can feasibly be covered within our GPU budget and reducing coverage of diverse WHAR settings (e.g., ExtraSensory [105] and Capture24 [18]). In addition, some datasets are distributed together with large auxiliary modalities such as video or audio that are tightly bundled and cannot be downloaded separately, resulting in even more prohibitive dataset sizes (e.g., Ego4D [35], CMU-MMAC [96], and ActionSense [28]).

Quality. We exclude datasets with quality issues, such as incomplete releases (e.g., SHL [33]), corrupted data (e.g., BMHAD [72]), or severe inconsistencies in activity annotations or substantial missing activity classes for individual subjects (e.g., KU-HAR [94]), as these issues hinder a meaningful evaluation.

Redundancy. Dataset extensions that only add modalities outside the scope of this work are treated as redundant to avoid unnecessary overlap. For example, Opportunity++ [24] is excluded, as it does not provide additional relevant information compared to OPPO (Locomotion) [85].

Citations. Finally, to ensure a minimum level of quality and adoption, we require datasets to have at least 30 citations. During dataset collection, we observed a large number of datasets with little to no citation footprint. Given the need to constrain the overall number of datasets in the benchmark, we prioritize those with demonstrated usage in the literature. The citation threshold thus serves as a proxy for practical relevance, indicating that a dataset has been used in prior WHAR research. While this may disadvantage very recent datasets, we found the threshold to be sufficiently permissive to still include newer datasets when their suitability for benchmarking is evident, e.g., WEAR [14]. Importantly, this criterion does not limit future extensions: newly published datasets that are suitable for WHAR benchmarking can be readily integrated into the WHAR Datasets library and used to construct customized benchmarks.

4.3 Dataset Suite

After applying the selection criteria, we obtain a final suite of 30 diverse datasets suitable for WHAR benchmarking. These datasets are summarized in Table 3, which provides an overview of their key characteristics, including publication year, citation count, number of subjects, number of activities, application settings, device types, sensor modalities, and number of channels. The resulting suite of datasets spans more than a decade of research, ranging from early datasets such as WISDM [51] and OPPO (Locomotion) [85] to more recent contributions such as WEAR [14] and Hang-Time [39]. The datasets exhibit

Table 3. Selected WHAR datasets, sorted by citation count, providing an overview of their key characteristics, including publication year, number of citations, subjects, activities, application settings, device types, sensor modalities, and number of channels.

Abbreviations: ADL=Activities of Daily Living, Loc=Locomotion, Fit=Fitness, Fall=Fall Detection, Acc=Accelerometer, Gyr=Gyroscope, Mag=Magnetometer, ECG=Electrocardiogram, EEG=Electroencephalogram, EMG=Electromyogram, GPS=Global Positioning System, Stretch=Stretch Sensor, Baro=Barometer, ALS=Ambient Light Sensor, Resp=Respiration Sensor, Gas=Gas Sensor, Flow=Airflow Sensor.

Dataset	Year	Citations	Subjects	Activities	Settings	Device Types	Sensor Modalities	Channels
WISDM [51]	2010	3862	36	6	ADL, Loc	Phone	Acc	3
UCI-HAR [6]	2013	3372	30	6	ADL, Loc	Phone	Acc, Gyr	9
ActRecTut (Gestures) [15]	2014	2086	2	11	ADL, Fit	Node	Acc, Gyr	15
PAMAP2 [82]	2012	1758	9	13	ADL, Loc, Fit	Node	Acc, Gyr, Mag, ECG	52
OPPO (Locomotion) [85]	2010	1024	4	18	ADL, Loc	Node	Acc, Gyr, Mag	133
HHAR [98]	2015	1019	9	6	ADL, Loc, Fit	Phone, Watch	Acc, Gyr	6
UTD-MHAD [19]	2015	997	8	27	ADL, Loc, Fit	Node	Acc, Gyr	6
HAPT [83]	2016	939	30	6	ADL, Loc	Phone	Acc, Gyr	6
MHEALTH [8]	2014	887	10	13	ADL, Loc, Fit, Health	Node	Acc, Gyr, Mag, ECG	23
DSADS [5]	2010	780	8	19	ADL, Loc, Fit	Node	Acc, Gyr, Mag	45
USC-HAD [114]	2012	753	14	12	ADL, Loc, Fit	Node	Acc, Gyr	6
SAD [90]	2014	752	10	7	ADL, Loc, Fit	Phone	Acc, Gyr, Mag	60
UniMiB-SHAR [63]	2017	712	30	17	ADL, Fall	Phone	Acc	3
Daphnet [64]	2009	652	10	3	Health, Loc	Node	Acc	9
RealWorld [101]	2016	482	15	8	ADL, Loc, Fit	Phone, Watch	Acc, Gyr, Mag, GPS, ALS	81
UP-Fall [60]	2019	462	17	11	ADL, Fall	Node, Helmet	Acc, Gyr, ALS, EEG	42
MotionSense [59]	2019	345	24	6	ADL, Loc	Phone	Acc, Gyr	18
UMAFall [17]	2017	243	19	15	ADL, Fall	Node, Phone	Acc, Gyr, Mag	39
REALDISP [9]	2014	216	17	34	ADL, Loc, Fit	Node	Acc, Gyr, Mag	117
RealLifeHAR [31]	2020	208	17	4	ADL, Loc	Phone	Acc, Gyr, Mag, GPS	15
WISDM-19 (Phone) [110]	2019	198	51	18	ADL, Loc	Phone	Acc, Gyr	6
WISDM-19 (Watch) [110]	2019	198	51	18	ADL, Loc	Watch	Acc, Gyr	6
HuGaDB [21]	2018	154	18	12	ADL, Loc, Fit	Node	Acc, Gyr, EMG	38
HARTH [54]	2021	132	22	12	ADL, Loc, Fit	Node	Acc	6
w-HAR [10]	2020	100	22	9	ADL, Loc	Node	Acc, Gyr, Mag, Stretch	7
WEAR [14]	2024	66	24	18	Fit	Watch	Acc	12
HAR70+ [7]	2021	55	18	7	ADL, Loc, Health	Node	Acc	6
Hang-Time [39]	2023	52	24	6	Fit	Watch	Acc	3
UCA-EHAR [71]	2022	35	20	12	ADL, Loc, Fit	Glasses	Acc, Gyr, Baro	7
GOTOV [76]	2022	33	31	16	ADL, Loc, Fit, Health	Node, Vest, Mask, Belt	Acc, ECG, Resp, Gas, Flow	28

substantial variation in scale, with the number of subjects ranging from 2 to 51 participants and the number of activities ranging from 3 to 34. This variability reflects differences in experimental design, task complexity, and target applications.

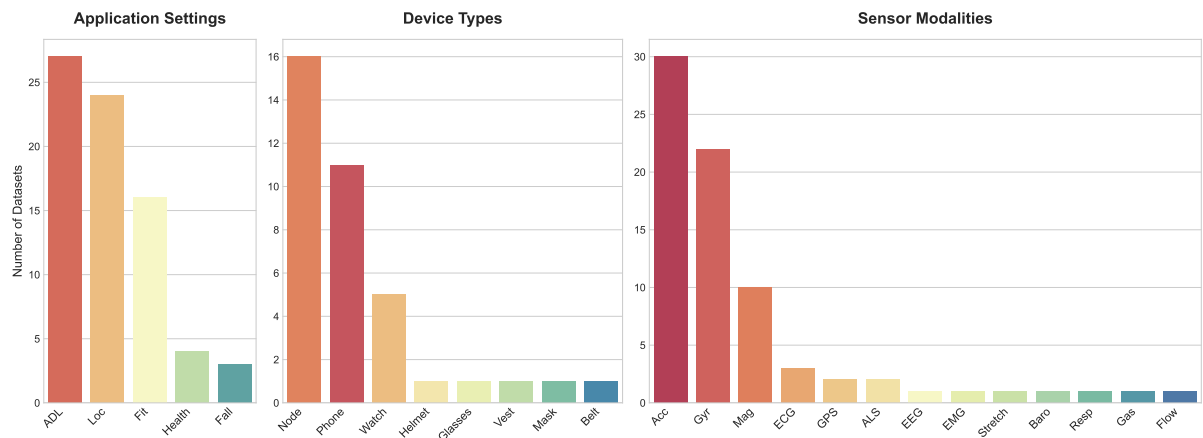


Fig. 3. A categorical overview of the 30 selected datasets. The bar charts highlight the heterogeneity of our dataset suite along different dimensions by detailing the frequency of application settings (left), utilized device types (center), and integrated sensor modalities (right). By intentionally including a broad spectrum of configurations, from single-sensor smartphone setups to highly multimodal sensor nodes, the suite enables a comprehensive evaluation of model generalizability across diverse real-world scenarios.

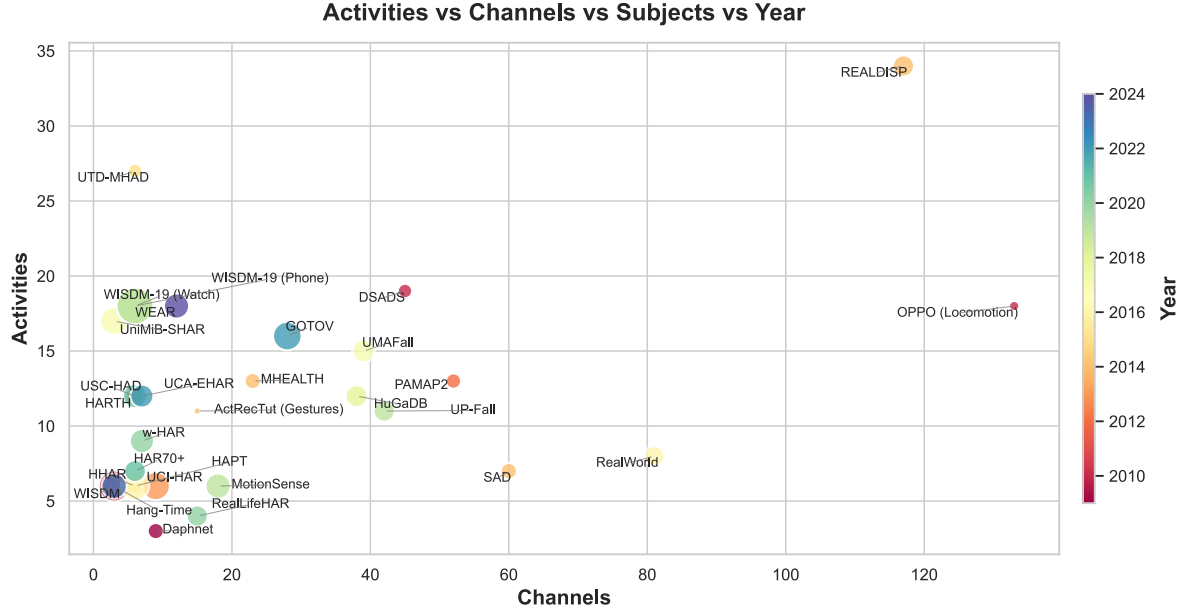


Fig. 4. A multi-dimensional overview of the 30 selected datasets. The chart highlights the diversity of the suite by mapping the number of sensor channels against the number of activity classes, with bubble size indicating the number of subjects and color encoding the publication year. The wide variation across these dimensions, including several outliers representing more extreme settings, enables our benchmark to assess the robustness of model architectures across a broad range of conditions.

In terms of application domains, the suite covers a wide spectrum of WHAR scenarios, including activities of daily living (ADL), locomotion, fitness tracking, fall detection, and health-related monitoring. Representative examples include UniMiB-SHAR [63] and UP-Fall [60] for fall detection, PAMAP2 [82] and DSADS [5] for fitness and daily activities, and WEAR [14] and Hang-Time [39] for fitness-specific scenarios.

The datasets further differ considerably in their sensor configurations. Many datasets are based on single-device setups, such as smartphones, smartwatches, or dedicated wearable nodes, typically equipped with inertial measurement units (IMUs) comprising accelerometer, gyroscope, and magnetometer signals. In contrast, other datasets employ multi-device or multi-modal configurations, including distributed body sensor networks or combinations of multiple wearable devices. Across the dataset suite, a wide range of sensor modalities is represented, including inertial sensors (accelerometer, gyroscope, magnetometer), physiological sensors (electrocardiogram (ECG), heart rate, electroencephalogram (EEG), electromyogram (EMG), respiration), environmental sensors (barometer, ambient light sensor), localization sensors (GPS), and specialized sensors such as stretch sensors. Consequently, the number of sensor channels varies widely across datasets, ranging from low-dimensional signals (e.g., 3 channels) to highly multivariate recordings exceeding 100 channels.

The heterogeneity of the dataset suite is a deliberate design choice. Rather than optimizing for a single application scenario, the benchmark aims to evaluate model robustness across diverse sensor modalities, activity domains, and data characteristics. This diversity enables a more comprehensive assessment of model performance and supports the identification of architectures that generalize well across varying real-world settings. In this sense, the benchmark is intended not only as a comparative tool but also as a practical reference for selecting suitable model architectures for specific tasks and sensor configurations.

4.4 WHAR Datasets Library

To address the comparability challenges in WHAR discussed in Section 2.2, we propose the open-source WHAR Datasets library. The current version ⁵ constitutes a fully released, substantially redesigned and expanded implementation and builds upon an earlier work-in-progress version that was introduced in a non-archival workshop paper [16]. The library provides a configuration-driven framework for standardizing data handling across heterogeneous WHAR datasets with the objective of improving reproducibility and comparability.

A central component of this framework is a session-centric standardized dataset format. A session is defined as the multivariate time series obtained from a continuous recording of a single subject performing a specific activity. To facilitate

⁵<https://github.com/teco-kit/whar-datasets>

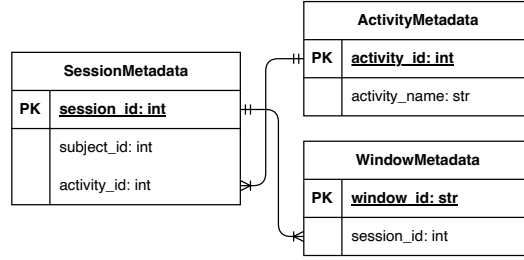


Fig. 5. Entity-relationship diagram illustrating the metadata schema for the standardized data format.

efficient data management, the framework separates raw sensor data from structural metadata. The metadata are organized into three centralized tabular files per dataset: activity metadata, session metadata, and window metadata, as illustrated in Figure 5. Activity metadata map numerical activity identifiers to descriptive labels. Session metadata associate each session identifier with an activity identifier. Window metadata record the identifiers of windows generated during preprocessing and link them to their corresponding session. This standardized representation enables a clear separation of concerns across four stages of data handling: preprocessing, split generation, postprocessing, and sample loading.

The preprocessing pipeline is executed once per configuration. It manages the downloading, extraction, and parsing of raw data into sessions, followed by dataset-specific activity and channel selection, resampling, and segmentation into windows. The exact behavior of each step is governed by a dataset-specific configuration file, which defines all preprocessing parameters as well as the corresponding parser required to interpret the raw data format. Intermediate results are cached using hash-based invalidation, which ensures deterministic reuse and allows selective recomputation when relevant configuration parameters change. The pipeline outputs the metadata tables as separate CSV files and stores the sensor data in two consolidated Parquet files, one containing all sessions and one containing all windows. This design avoids the proliferation of many small files, which can lead to file-system limitations related to inode counts and increased memory overhead.

Following preprocessing, the framework provides a set of splitter classes that partition windows into training, validation, and test sets. Each splitting strategy is implemented as a dedicated splitter class, enabling a clear and modular representation of different evaluation protocols. These splitters operate independently of preprocessing, allowing protocols to be modified without reprocessing the dataset. Splitting is performed exclusively on metadata, without direct access to raw sensor data. The library includes implementations for common strategies such as leave-one-subject-out (LOSO), k-fold, and k-subject-groups (KSG) cross-validation, and can be readily extended with custom splitter classes. Each splitter incorporates strict overlap checks to prevent data leakage across partitions.

The postprocessing pipeline performs normalization and optional feature transformations. It is intentionally decoupled from preprocessing in order to operate in a split-specific manner, as normalization depends on statistics derived from the training data. Accordingly, normalization parameters are computed exclusively on the training partition and subsequently applied consistently to all partitions. This design enables efficient reuse across different evaluation protocols. For example, in a LOSO cross-validation setting, each split defined by a different held-out subject requires only the execution of the postprocessing pipeline, without rerunning the preprocessing pipeline. In addition, configurable transformations, such as Fourier or short-time Fourier transforms, can be applied to the windowed sensor data. The resulting samples are cached to enable efficient reuse in subsequent experiments. All postprocessing steps and parameters are defined within the dataset-specific configuration file.

For model training, the library provides a framework-agnostic sample loader with adapters for common deep learning frameworks such as PyTorch [77] and TensorFlow [1]. The loader binds metadata to stored samples, enabling efficient retrieval and filtering by subject or activity. Depending on the configuration and available hardware resources, samples can be preloaded into memory or accessed on demand. The loader can also compute class weights from window metadata to address class imbalance during optimization.

Using this modular architecture, all datasets in the curated benchmark suite described in Table 3 were successfully integrated into the WHAR Datasets library, establishing a large, harmonized repository for comprehensive benchmarking. This integration demonstrates the library’s ability to accommodate diverse data formats, sampling rates, and sensor configurations. The design further supports extensibility by allowing researchers to incorporate new datasets through dataset-specific parsers and configuration files. This approach enables the broader research community to adopt a unified standardization and evaluation pipeline without modifying the core system, thereby supporting more consistent and comparable WHAR research.

5 Benchmarking

Having defined the model and dataset suites (see [Section 3.3](#) and [Section 4.3](#)) and standardized them using the WHAR Models library (see [Section 3.4](#)) and the WHAR Datasets library (see [Section 4.4](#)), we now proceed to benchmarking. We first introduce the benchmarking protocol in [Section 5.1](#), followed by large-scale experiments conducted under this protocol, with cross-subject predictive performance reported in [Section 5.2](#). We also conduct on-device evaluations to measure deployment efficiency metrics in [Section 5.3](#) and analyze their trade-offs with predictive performance in [Section 5.4](#), where Pareto frontiers are established. Finally, we define a relative ranking to construct leaderboards across different deployment efficiency metrics in [Section 5.5](#) and derive practical insights to guide researchers in selecting suitable model architectures for specific WHAR tasks, as discussed in [Section 5.6](#).

5.1 Benchmarking Protocol

All experiments are conducted using a unified benchmarking protocol applied consistently across models and datasets. For the datasets, this is implemented by defining a configuration file for each dataset within the WHAR Datasets library using the specified hyperparameters. Preprocessing is configured to perform windowing with a window size of 3 s with 50 % overlap while preserving each dataset’s native sampling rate. The postprocessing pipeline applies global standardization, where normalization parameters are estimated exclusively from the train set and then applied to all partitions. For data splitting, we use a k -subject-groups (KSG) splitter with $k = 10$. Leave-one-subject-out cross-validation is not used due to the large number of resulting splits and the associated GPU resource constraints, although it would likely yield slightly better predictive performance since more training data would be available in each split. In each split, the samples from the held-out group of subjects define the test set, while the remaining samples are divided into train and validation sets, with the validation set comprising 20 % of the data. For each dataset and each split, the 14 deep model architectures are trained using the AdamW [55] optimizer, configured with a cosine-annealing learning rate of 10^{-3} , weight decay 10^{-4} , batch size of 64, and a maximum of 100 epochs. We apply early stopping with a patience of 10 based on macro-F1 score on the validation set, after which the best validation checkpoint is restored for model selection before final evaluation. A cross-entropy loss function is used, weighted by class frequencies computed from the train set to mitigate class imbalance. For the three classical baselines, we extract tsfresh [23] features using MinimalFCParameters and train scikit-learn [78] implementations of KNeighborsClassifier with $k = 5$ and distance weighting, RandomForestClassifier with 100 trees, Gini split criterion, maximum depth 8, minimum leaf size 10, and no cost-complexity pruning, and SVC with an RBF kernel, $C = 1.0$, and $\gamma = \text{scale}$; all unspecified hyperparameters use the scikit-learn defaults.

5.2 Predictive Performance Evaluation

Final predictive performance is reported exclusively on the test set containing unseen subjects, reflecting realistic deployment conditions under inter-subject variability. Predictive performance is evaluated using the macro-F1 score, which assigns equal importance to all classes to ensure balanced results across both minority and majority activity classes. For each model–dataset combination, we report the mean and standard deviation over all splits. This corresponds to a cell in the model–dataset predictive performance matrix visualized in [Figure 6](#). [Figure 7](#) complements this matrix with two aggregate views, reporting each model’s mean/min/max test macro-F1 across datasets, as well as each dataset’s mean/min/max across models. Together, these views separate model robustness (how often models fail on challenging datasets) from dataset sensitivity (how strongly outcomes depend on the choice of architecture).

[Figure 6](#) shows that there is no single clear winner model across all datasets. At the aggregate level, the strongest models are tightly clustered rather than separated by large margins. CNN-HAR attains the highest mean test macro-F1 (67.7%), followed very closely by TinyHAR (67.6%) and TripleCrossDomainAttention (67.6%), with several additional models remaining near this group. These small differences should not be overinterpreted as clear evidence of superiority. Instead, they indicate that benchmark-level performance is distributed across a set of closely competing architectures rather than dominated by one universally superior model. The per-dataset winners further emphasize this heterogeneity. Across the full 30 datasets, RandomForest leads 9 times, CNN-HAR and DynamicWHAR 5 each, DANA 3, TripleCrossDomainAttention and TinyHAR 2 each, and TinierHAR, MLP Mixer, MLP HAR, and Guan-LSTM 1 each. Notably, CNN-HAR has the highest aggregate mean score but wins only five individual datasets, indicating that its advantage stems from consistently strong performance rather than clear dominance on a subset of benchmarks. Taken together, the orange winner cells in [Figure 6](#) show that no single architecture performs best across all datasets.

Dataset	sad	95.60 ±5.91%	97.67 ±5.22%	90.77 ±8.68%	93.41 ±10.89%	91.22 ±12.37%	95.68 ±7.20%	95.67 ±5.77%	98.25 ±1.38%	91.43 ±12.94%	97.38 ±2.75%	91.14 ±11.23%	93.51 ±8.45%	90.10 ±13.00%	95.45 ±5.80%	81.24 ±24.82%	90.36 ±11.30%	82.09 ±11.58%	92.41 ±9.37%
	uci_har	96.03 ±2.91%	96.11 ±2.08%	95.96 ±2.63%	94.97 ±2.56%	95.17 ±3.81%	94.05 ±4.35%	92.12 ±4.03%	84.69 ±5.26%	92.57 ±4.64%	93.19 ±3.74%	93.00 ±4.14%	94.57 ±6.78%	91.46 ±4.94%	90.02 ±5.09%	92.27 ±6.62%	72.72 ±6.26%	71.23 ±5.39%	90.60 ±4.26%
	motion_sense	91.51 ±6.98%	91.20 ±6.47%	91.34 ±4.02%	88.91 ±6.94%	90.90 ±5.40%	88.57 ±6.31%	93.48 ±4.15%	91.49 ±3.67%	88.48 ±6.68%	91.16 ±5.30%	87.56 ±7.52%	91.24 ±5.92%	90.52 ±7.84%	88.60 ±6.23%	85.26 ±9.47%	76.54 ±11.51%	67.42 ±8.77%	87.89 ±6.66%
	wear	86.90 ±8.45%	86.48 ±8.32%	88.52 ±6.60%	87.01 ±7.66%	87.04 ±9.04%	85.63 ±9.32%	89.65 ±5.42%	87.34 ±5.55%	84.16 ±8.84%	85.28 ±9.15%	83.33 ±10.31%	83.99 ±10.64%	82.81 ±11.89%	81.13 ±8.18%	84.24 ±10.61%	75.92 ±10.27%	69.67 ±10.65%	84.06 ±8.88%
	dsads	86.87 ±6.74%	86.77 ±6.61%	85.57 ±7.83%	84.42 ±7.11%	86.40 ±7.01%	86.15 ±4.18%	87.85 ±6.44%	89.33 ±4.03%	83.79 ±9.80%	84.26 ±8.04%	81.98 ±9.31%	81.50 ±5.18%	83.22 ±6.90%	85.69 ±7.01%	77.43 ±8.46%	75.88 ±6.44%	70.94 ±4.93%	83.41 ±6.82%
	mhealth	86.13 ±5.16%	80.53 ±10.16%	81.28 ±8.26%	82.96 ±6.97%	84.76 ±6.45%	83.95 ±9.31%	85.74 ±5.58%	87.65 ±4.83%	80.52 ±9.34%	80.76 ±7.20%	78.03 ±9.72%	77.20 ±10.70%	78.44 ±6.96%	81.91 ±6.12%	76.50 ±12.08%	76.61 ±10.21%	64.17 ±16.66%	80.42 ±8.57%
	wisdm	81.44 ±9.10%	85.00 ±6.17%	83.55 ±4.66%	85.49 ±5.01%	83.77 ±5.56%	83.79 ±5.61%	81.93 ±7.40%	71.58 ±6.52%	84.65 ±5.52%	76.07 ±4.53%	83.99 ±4.82%	82.98 ±5.77%	84.24 ±4.18%	82.97 ±4.97%	65.06 ±7.59%	59.16 ±4.44%	51.19 ±6.90%	78.05 ±5.81%
	real_disp	86.97 ±10.93%	88.30 ±10.07%	87.16 ±9.48%	84.08 ±10.85%	86.26 ±11.17%	86.39 ±11.28%	68.96 ±13.93%	84.91 ±8.42%	80.07 ±12.13%	85.41 ±10.57%	79.35 ±12.87%	66.11 ±15.19%	68.31 ±15.57%	80.22 ±10.64%	80.68 ±11.45%	52.85 ±14.30%	38.31 ±9.37%	76.73 ±11.66%
	hapt	84.80 ±5.58%	81.02 ±6.79%	82.02 ±6.06%	83.58 ±6.23%	81.42 ±6.23%	81.38 ±6.09%	79.41 ±5.66%	67.13 ±5.50%	79.52 ±6.66%	79.22 ±6.11%	77.33 ±7.17%	80.50 ±5.76%	73.44 ±10.25%	59.74 ±5.33%	75.69 ±8.68%	63.86 ±6.59%	66.34 ±7.96%	76.26 ±6.61%
	hhar	82.47 ±8.73%	80.67 ±8.23%	78.06 ±8.06%	80.02 ±10.17%	78.59 ±8.17%	79.95 ±8.08%	78.82 ±7.90%	75.38 ±10.37%	75.39 ±8.65%	72.96 ±10.92%	77.69 ±9.56%	80.33 ±8.23%	75.04 ±11.66%	77.91 ±10.89%	79.21 ±9.63%	65.16 ±10.96%	47.60 ±10.94%	75.60 ±9.48%
	usc_had	76.16 ±14.75%	78.09 ±8.48%	74.34 ±6.32%	71.11 ±21.63%	75.67 ±16.42%	73.32 ±19.35%	75.70 ±27.27%	72.68 ±7.53%	70.99 ±11.07%	76.52 ±11.41%	73.50 ±8.87%	78.83 ±10.66%	68.88 ±9.11%	62.88 ±11.16%	69.82 ±11.83%	37.74 ±10.09%	43.25 ±12.37%	69.38 ±14.32%
	wisdm_19_watch	68.79 ±4.62%	68.72 ±4.42%	68.51 ±5.04%	71.30 ±4.65%	68.60 ±4.97%	69.38 ±4.42%	68.93 ±4.33%	59.80 ±5.50%	69.04 ±4.36%	61.80 ±4.46%	67.18 ±5.63%	69.77 ±5.33%	70.61 ±5.14%	67.82 ±4.95%	65.39 ±5.20%	49.76 ±5.44%	43.91 ±4.51%	65.25 ±4.88%
	opportunity	63.12 ±14.03%	65.37 ±8.48%	70.57 ±6.32%	58.70 ±21.63%	65.88 ±16.42%	61.44 ±19.35%	62.65 ±27.27%	70.63 ±7.53%	65.20 ±11.07%	74.71 ±11.41%	62.15 ±8.87%	60.26 ±14.75%	65.88 ±11.34%	65.14 ±12.28%	58.08 ±18.83%	66.68 ±10.09%	59.42 ±12.37%	64.46 ±14.32%
	harth	72.38 ±7.09%	71.14 ±7.36%	72.69 ±8.56%	69.86 ±7.44%	72.64 ±8.26%	66.68 ±6.48%	67.53 ±6.53%	55.90 ±4.99%	63.38 ±5.52%	65.15 ±6.63%	64.03 ±8.27%	66.84 ±6.94%	66.72 ±10.30%	54.07 ±6.56%	61.54 ±8.70%	44.28 ±3.90%	43.06 ±4.73%	63.40 ±7.02%
	gotov	64.38 ±14.83%	66.27 ±13.06%	68.02 ±12.25%	66.81 ±13.58%	64.93 ±12.35%	65.16 ±12.91%	70.09 ±14.47%	65.63 ±14.27%	64.24 ±13.15%	64.40 ±13.47%	63.27 ±14.76%	59.92 ±13.21%	60.05 ±15.49%	59.70 ±14.73%	59.94 ±14.70%	51.37 ±11.84%	44.45 ±9.42%	62.27 ±13.44%
	real_world	57.55 ±19.04%	60.18 ±15.86%	61.69 ±16.81%	63.97 ±19.06%	60.89 ±17.74%	57.20 ±21.19%	68.39 ±20.77%	80.35 ±16.33%	59.25 ±20.23%	58.86 ±10.79%	55.03 ±15.71%	51.85 ±12.84%	50.80 ±19.52%	67.26 ±21.81%	52.97 ±20.02%	56.72 ±18.67%	47.29 ±12.10%	59.43 ±17.56%
	har70	69.24 ±9.08%	65.80 ±11.32%	59.50 ±9.08%	60.60 ±9.80%	68.47 ±10.13%	62.36 ±8.13%	57.49 ±8.75%	57.50 ±9.28%	59.43 ±7.33%	59.84 ±11.02%	60.37 ±9.06%	62.42 ±10.37%	56.24 ±7.90%	53.31 ±6.83%	56.33 ±6.16%	48.05 ±6.20%	48.97 ±8.46%	57.98 ±8.76%
	pamap2	62.06 ±23.13%	62.20 ±22.44%	61.94 ±23.13%	60.27 ±22.07%	59.63 ±20.95%	62.25 ±21.94%	64.24 ±24.58%	65.06 ±23.96%	57.58 ±21.67%	63.09 ±23.86%	55.81 ±24.08%	47.81 ±20.33%	51.82 ±20.60%	55.91 ±22.20%	58.97 ±21.76%	48.05 ±24.97%	48.97 ±23.48%	57.98 ±22.66%
	up_fall	55.02 ±9.35%	60.96 ±8.14%	58.66 ±8.40%	52.62 ±7.88%	55.65 ±6.97%	54.49 ±7.70%	61.12 ±7.68%	72.77 ±4.27%	52.24 ±8.95%	56.54 ±4.58%	52.87 ±9.51%	43.16 ±9.90%	45.67 ±9.98%	54.53 ±6.88%	51.73 ±7.35%	46.85 ±11.79%	40.43 ±6.99%	53.84 ±8.07%
	utd_mhad	68.16 ±13.42%	65.75 ±12.12%	66.97 ±12.52%	64.80 ±12.98%	61.00 ±11.75%	67.21 ±12.48%	64.66 ±13.01%	51.20 ±11.38%	59.40 ±9.65%	61.59 ±10.10%	54.91 ±7.51%	64.90 ±8.78%	24.69 ±12.12%	19.02 ±4.72%	60.96 ±11.25%	23.92 ±3.23%	33.85 ±7.06%	53.71 ±10.24%
	uca_ear	58.66 ±6.58%	57.97 ±7.39%	58.09 ±8.28%	60.10 ±8.39%	55.99 ±6.73%	59.85 ±9.39%	57.36 ±7.45%	47.59 ±3.24%	56.83 ±9.34%	53.00 ±6.71%	56.33 ±6.37%	57.87 ±8.00%	59.47 ±8.50%	42.85 ±5.12%	50.35 ±7.93%	36.68 ±6.65%	31.73 ±6.03%	52.98 ±7.18%
	unimib_shar	63.30 ±8.00%	58.65 ±9.70%	55.22 ±5.55%	62.42 ±8.59%	58.41 ±6.60%	60.97 ±7.19%	52.51 ±6.03%	44.49 ±5.83%	61.08 ±6.51%	42.26 ±5.83%	56.34 ±6.51%	57.19 ±8.39%	55.55 ±13.38%	37.95 ±5.07%	38.71 ±5.31%	19.20 ±2.54%	23.45 ±3.20%	49.86 ±7.09%
	real_life_har	47.64 ±9.62%	49.05 ±9.25%	49.33 ±9.11%	52.56 ±9.93%	48.16 ±8.69%	53.05 ±7.60%	55.01 ±8.60%	62.56 ±17.66%	48.72 ±10.33%	46.30 ±9.54%	48.11 ±7.95%	42.30 ±11.15%	43.32 ±11.29%	53.59 ±10.84%	42.85 ±11.06%	48.70 ±10.44%	38.04 ±10.65%	48.78 ±10.22%
	hang_time	51.50 ±1.81%	51.14 ±1.87%	51.08 ±2.53%	52.15 ±1.87%	50.93 ±2.43%	50.95 ±1.80%	46.98 ±2.05%	46.94 ±2.58%	51.49 ±2.00%	47.78 ±2.37%	51.50 ±2.34%	50.79 ±2.64%	52.29 ±2.55%	46.49 ±2.05%	42.90 ±3.32%	36.77 ±1.96%	39.19 ±1.60%	48.29 ±2.22%
	w_har	52.72 ±3.39%	52.16 ±4.90%	49.83 ±6.49%	49.31 ±7.78%	49.14 ±8.90%	48.63 ±7.83%	49.91 ±8.90%	53.07 ±4.20%	48.81 ±7.72%	46.28 ±6.88%	47.23 ±6.78%	48.88 ±8.56%	39.09 ±11.06%	35.78 ±4.56%	50.97 ±4.66%	31.48 ±6.58%	32.43 ±1.52%	46.22 ±2.68%
	uma_fall	49.99 ±13.96%	50.17 ±10.84%	52.43 ±11.66%	51.85 ±13.76%	51.22 ±12.26%	50.60 ±13.31%	56.18 ±12.60%	54.19 ±16.09%	45.71 ±11.64%	49.06 ±14.12%	45.49 ±12.73%	33.05 ±10.77%	37.31 ±12.42%	40.82 ±12.13%	41.88 ±12.36%	29.76 ±13.01%	29.19 ±11.10%	45.23 ±12.63%
	hugadb	49.15 ±7.73%	46.63 ±6.73%	47.79 ±9.10%	44.82 ±8.85%	48.26 ±8.73%	42.94 ±8.83%	44.14 ±7.84%	52.47 ±5.65%	42.86 ±7.31%	46.14 ±8.33%	44.31 ±7.83%	41.94 ±7.48%	44.55 ±8.71%	43.88 ±7.17%	41.42 ±8.57%	41.22 ±6.97%	38.48 ±7.23%	44.76 ±7.83%
	actrectut_gestures	52.25 ±9.71%	51.06 ±7.00%	61.49 ±0.07%	57.23 ±8.32%	45.17 ±7.33%	53.76 ±15.68%	41.47 ±11.83%	41.89 ±18.12%	52.53 ±9.07%	43.78 ±0.58%	48.21 ±5.22%	37.77 ±4.54%	35.42 ±0.27%	46.89 ±1.44%	31.07 ±6.27%	27.59 ±2.60%	28.64 ±2.35%	44.48 ±6.49%
	daphnet	40.23 ±7.73%	43.94 ±7.31%	42.72 ±6.59%	43.59 ±9.33%	45.41 ±9.52%	43.01 ±5.69%	37.81 ±5.34%	33.84 ±10.37%	41.81 ±10.37%	40.63 ±6.30%	40.66 ±6.12%	38.93 ±7.48%	29.20 ±6.73%	35.44 ±4.66%	37.69 ±6.58%	31.73 ±1.52%	31.11 ±2.34%	38.69 ±2.58%
	wisdm_19_phone	30.45 ±4.28%	29.17 ±3.54%	32.05 ±2.43%	31.72 ±3.69%	30.28 ±2.68%	29.53 ±5.34%	32.23 ±3.41%	26.81 ±2.20%	29.07 ±4.50%	23.60 ±3.35%	27.64 ±4.58%	27.31 ±4.17%	26.85 ±6.17%	31.68 ±2.82%	23.85 ±4.23%	18.46 ±2.59%	17.59 ±4.16%	27.55 ±3.77%
	Mean	67.72 ±8.94%	67.61 ±8.62%	67.57 ±8.01%	67.02 ±9.41%	66.73 ±8.85%	66.61 ±9.40%	66.27 ±9.21%	65.10 ±8.05%	64.67 ±9.35%	64.23 ±7.69%	63.61 ±9.31%	62.46 ±8.81%	60.07 ±9.88%	59.96 ±8.13%	59.84 ±10.11%	50.15 ±8.41%	46.25 ±8.40%	62.70 ±8.86%
		CNN-HAR	TinyHAR	Triple-Cross-Attn	DANA	TiniHAR	Attend+Discriminate	DynamicWHAR	Random Forest	DeepConvLSTM-Attn	MLP-HAR	DeepConvLSTM	MLP-Mixer	Guan-LSTM	DeepConvShallowLSTM	SA-HAR	SVM	k-NN	Mean
		Model																	

Fig. 6. Model–dataset predictive performance matrix with datasets as rows and models as columns. Each cell reports the mean test macro-F1 over all completed splits. The bottom row shows the mean macro-F1 of each model across datasets, while the rightmost column shows the mean for each dataset across models. Both axes are ordered by these means: models are sorted by their average performance across datasets, and datasets by their average performance across models. For each dataset, the best-performing model is highlighted in orange. The close clustering of mean macro-F1 scores indicates that no single model consistently dominates across all datasets.

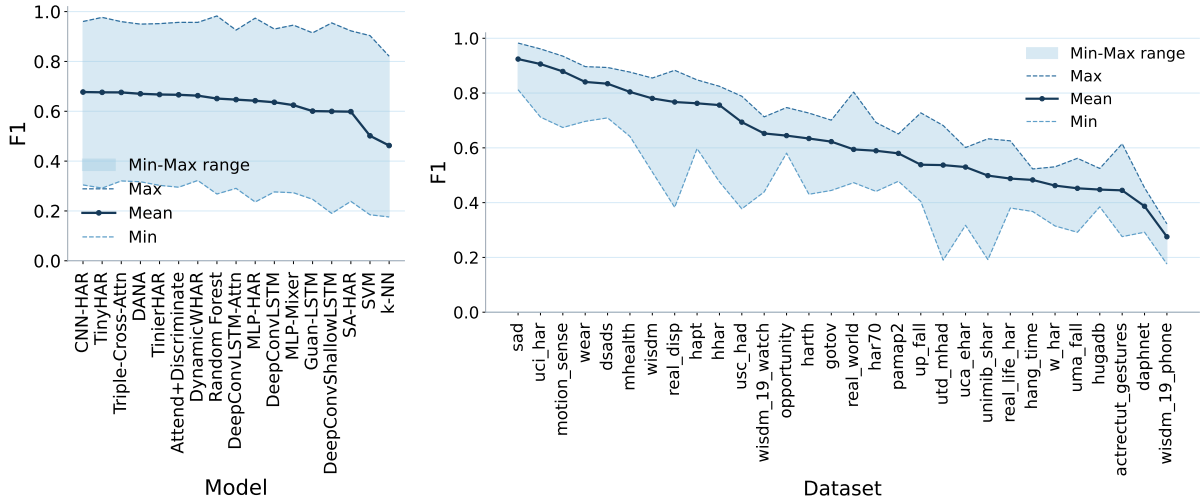


Fig. 7. Aggregate line-and-range summary plots derived from Figure 6. The per-model summary (left) reports the mean, min, and max test macro-F1 across all datasets, while the per-dataset summary (right) reports the same statistics across all evaluated models.

At the model level as seen in Figure 7, most leading deep model architectures and RandomForest occupy a very similar band across all three summary statistics (mean, min, max) over datasets. This reinforces that, among competitive architectures, small differences in mean test macro-F1 should be interpreted cautiously. The clearest drop-off appears for KNN and SVM, whose performance bands lie well below the main group. This pattern suggests that, among competitive model architectures, test macro-F1 alone is often insufficient as the primary model-selection criterion. At the dataset level, as shown in Figure 7, difficulty remains highly heterogeneous. SAD (92.4% mean), UCI HAR (90.6%), and MotionSense (87.9%) are close to saturation. Among the consistently difficult benchmarks, WISDM-19-Phone remains the hardest (27.5% mean, 32.2% best), followed by Daphnet (38.7% mean, 45.4% best). Range width further highlights where architecture choice matters most: RealDisp (49.99 points), UTD-MHAD (49.14 points), and UniMiB-SHAR (44.10 points) exhibit the largest gaps between worst and best models, while HUGADB (13.99 points), WISDM-19 Phone (14.64 points), and Hang Time (15.52 points) have comparatively narrow ranges, indicating settings that are either uniformly difficult or only weakly sensitive to architecture choice.

Pairwise rank correlations across datasets quantify this transferability. The mean Spearman correlation between dataset-specific model rankings is 0.485 across 870 off-diagonal dataset pairs, but values range from strongly aligned (UMA-Fall vs. WEAR: $\rho = 0.958$) to clearly conflicting (HHAR vs. Opportunity: $\rho = -0.426$). This variability highlights that benchmark-wide mean scores are useful summaries, but that model performance remains strongly dataset-dependent and should not be inferred from single-dataset wins alone.

5.3 Deployment Efficiency Evaluation

For deployment on wearable and edge devices, predictive performance alone is insufficient, as models must also satisfy strict hardware constraints. We therefore characterize deployment efficiency along three complementary dimensions: inference latency, which determines whether predictions can be computed in real time, peak memory overhead, which captures the maximum additional memory required during inference excluding the model itself, and exported model size, which reflects the memory required to store and load the model into memory. We additionally estimate approximate forward-pass FLOPs as a hardware-agnostic measure of computational cost, reported in Section A.3.

Inference latency and peak memory overhead are measured using a dedicated Android benchmarking app that executes each exported model directly on a Google Pixel 8 smartphone via ExecuTorch [79]. For each model-dataset combination we perform one warm-up pass and 50 repeated inference runs. Latency is reported in Section A.1 as the average time per prediction in milliseconds. Peak memory overhead is estimated using the app’s proportional set size (PSS), and we report the maximum increase relative to the pre-inference baseline in megabytes in Section A.2. These measurements reveal variability that is not visible in model-level averages: while some datasets preserve the overall ranking of models, others deviate in ways that are relevant for deployment decisions. Dataset-specific results are therefore essential when targeting a fixed WHAR

application setting or device budget. However, when selecting default deployment candidates, the trade-off between predictive performance and deployment efficiency across architectures must be considered.

5.4 Performance-Efficiency Trade-Offs

Beyond absolute efficiency, it is more informative to examine how deployment cost relates to predictive performance. To this end, we plot each efficiency metric against test macro-F1 in Figure 8, using mean values across datasets. CNN-HAR achieves the highest mean macro-F1 (67.7%) while remaining deployment-feasible, with 11.28 ms latency, 4.20 MB peak memory overhead, and a 0.95 MB model size. TinyHAR offers nearly identical accuracy (67.6%) with slightly lower latency (9.60 ms) but higher memory usage (6.34 MB) and a 1.02 MB model size, while TripleCrossDomainAttention matches this accuracy at 12.48 ms but incurs higher memory (8.15 MB) and model size (3.10 MB). TinierHAR stands out as the most compact strong neural model, maintaining 66.7% macro-F1 with only 4.30 ms latency, 1.06 MB memory overhead, and a 0.11 MB model size. In contrast, heavier recurrent and hybrid models are significantly more expensive without delivering predictive performance gains.

To identify the best trade-offs, we compute Pareto fronts for each metric and illustrate them as lines in Figure 8. A model lies on the Pareto front if no other model is both more accurate and more efficient under the same cost metric. For latency versus macro-F1, the Pareto front includes RandomForest, TinierHAR, TinyHAR, and CNN-HAR, forming a smooth progression from simple low-latency baselines to stronger compact neural model architectures. For peak memory and model size, the Pareto fronts are narrower and consist only of TinierHAR and CNN-HAR. Across all efficiency dimensions, these two models consistently provide the strongest trade-offs.

5.5 Leaderboards for Relative Rankings

While Figure 8 provides insight into which model architectures are practically useful and which are not, it does not provide a relative ranking that can be used to construct a leaderboard for the state of the art in WHAR that jointly accounts for predictive performance and deployment efficiency. To address this, we introduce a metric that relates the mean test macro-F1 score, averaged across all datasets, to the mean deployment cost, also averaged across datasets. The deployment cost is defined in terms of latency L , peak memory overhead M , or model size S .

To obtain a relative ranking, we first apply min-max normalization. For a metric x and model i , this is defined as:

$$x_{\text{norm},i} = \frac{x_i - \min_j(x_j)}{\max_j(x_j) - \min_j(x_j)}. \quad (1)$$

Based on these normalized quantities, we define a relative ranking metric. The deployment cost θ is first log-transformed as $\theta_i = \log(c_i)$, where c_i denotes the raw deployment cost prior to normalization. This transformation reduces the large dynamic range between compact and heavy models.

Each model is then scored according to its Euclidean distance to an ideal point that represents maximum normalized accuracy and minimum normalized cost, with equal weighting assigned to both components. The ideal point is given by $(F1_{\text{norm}}, \theta_{\text{norm}}) = (1, 0)$. The resulting score, referred to as the Efficiency-Index, is defined as:

$$E_i = 1 - \sqrt{0.5(1 - F1_{\text{norm},i})^2 + 0.5\theta_{\text{norm},i}^2}. \quad (2)$$

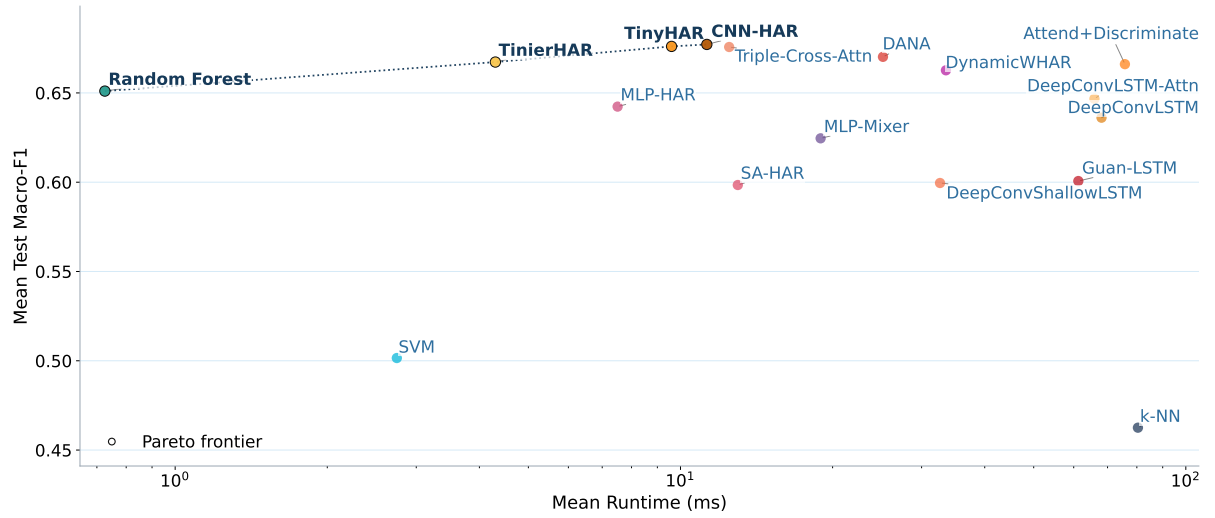
A score of 1 indicates that a model achieves both the highest normalized macro-F1 and the lowest normalized deployment cost. A score of 0 corresponds to the opposite extreme, that is, the lowest normalized macro-F1 and the highest normalized deployment cost. Higher values therefore indicate models that are closer to the desirable regime of high accuracy and low deployment cost, rather than models that optimize only a raw performance-to-cost ratio.

To construct a leaderboard that accounts for all three deployment costs, we further define a Joint-Efficiency-Index:

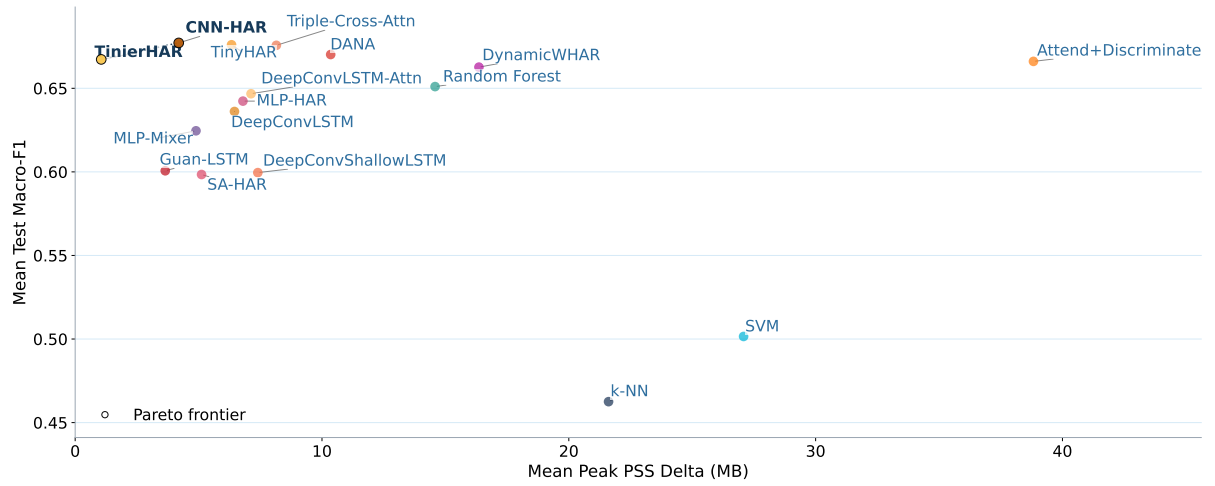
$$E_i^{\text{joint}} = 1 - \sqrt{0.5(1 - F1_{\text{norm},i})^2 + \frac{1}{6}L_{\text{norm},i}^2 + \frac{1}{6}M_{\text{norm},i}^2 + \frac{1}{6}S_{\text{norm},i}^2}, \quad (3)$$

where $L_{\text{norm},i}$, $M_{\text{norm},i}$, and $S_{\text{norm},i}$ denote the min-max-normalized log-latency, log-peak-memory, and log-model-size, respectively. This combined score assigns a weight of 0.5 to macro-F1 and distributes the remaining 0.5 equally across latency, peak memory overhead, and model size. The resulting leaderboard is shown in the fourth panel of Figure 9.

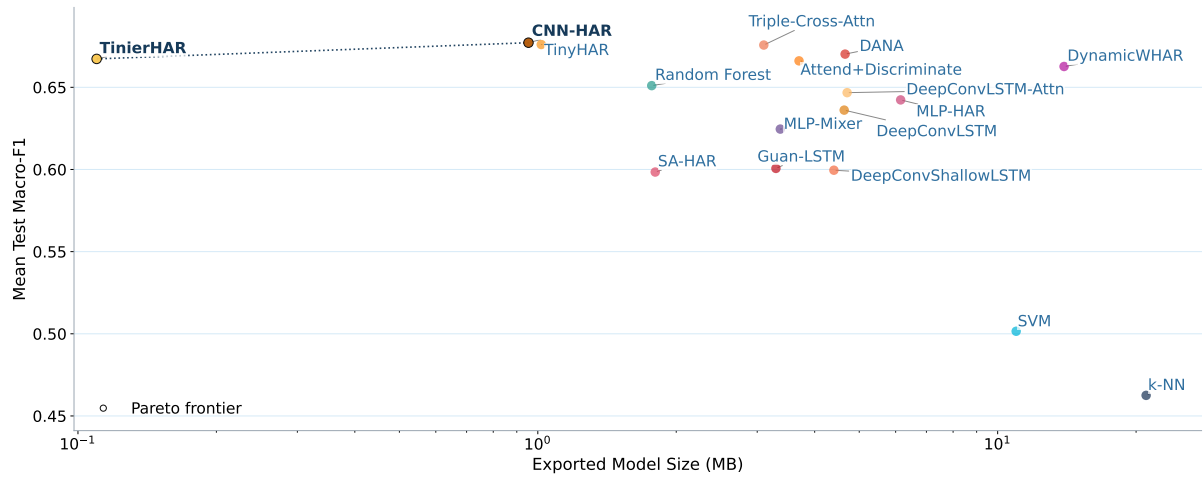
For each deployment efficiency metric, we compute these scores to obtain relative rankings, which are presented as leaderboards in the first three panels of Figure 9. These rankings closely reflect the Pareto fronts discussed in Section 5.4, with Pareto-optimal models generally occupying the highest positions. The only exception is MLP-HAR, which ranks slightly ahead of TinyHAR and CNN-HAR in the latency-based leaderboard. Overall, these leaderboards provide a concise one-dimensional



(a) Latency versus macro-F1 Pareto trade-off.



(b) Memory versus macro-F1 Pareto trade-off.



(c) Model size versus macro-F1 Pareto trade-off.

Fig. 8. Pareto-front deployment trade-off summaries for benchmark models with available runtime measurements and exported artifacts. The three panels report mean deployment cost versus mean test macro-F1 for latency, peak-memory overhead, and exported model size, respectively.

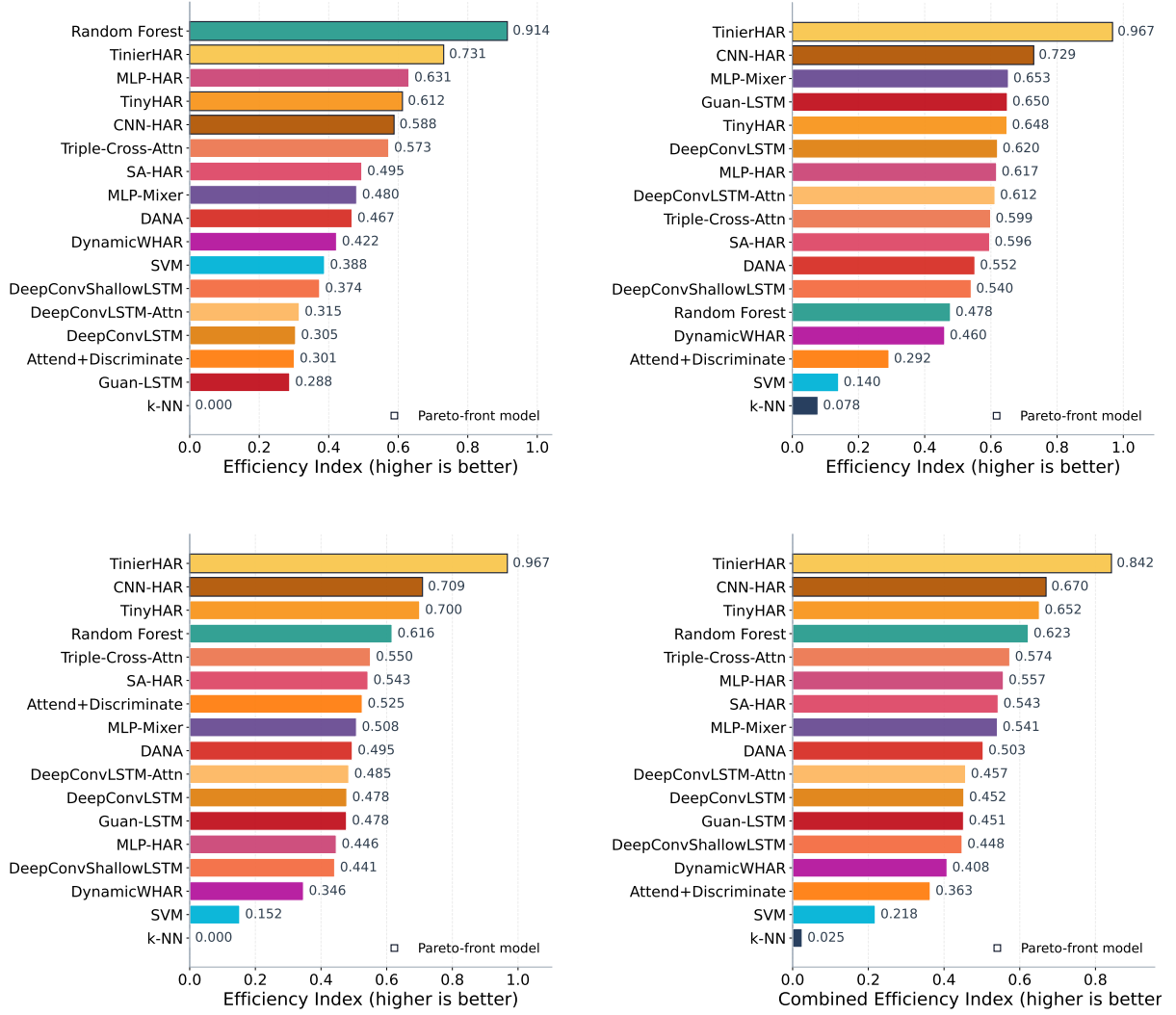


Fig. 9. Efficiency-index rankings for the same deployment dimensions shown in Figure 8. The first three panels summarize the aggregated performance-to-cost ratio under latency, peak-memory overhead, and exported model size, respectively. The fourth panel combines all three deployment costs with macro-F1 using weight 0.5 for macro-F1 and weight 1/6 for each cost term. Higher values indicate better efficiency.

summary of model performance under different deployment constraints. This representation facilitates the selection of suitable model architectures for specific application requirements and deployment conditions.

5.6 Practical Takeaways

From a practical perspective three main insights emerge. First, compact neural model architectures dominate the practically useful deployment region, while heavier recurrent and hybrid models tend to incur additional cost without consistent predictive performance gains. Second, RandomForest remains the strongest classical baseline: it defines the lowest-latency extreme and achieves competitive predictive performance, but it lags behind the most compact neural models on both peak memory and model size. Third, there is no universally optimal model. Instead, the choice depends on deployment constraints. TinierHAR is the safest option under strict memory or model size limits, while CNN-HAR provides the strongest overall trade-off between predictive performance and deployment efficiency.

6 Discussion

Our large-scale benchmark addresses the comparability crisis outlined in Section 1 and Section 2 by holding the data processing pipeline, cross-subject evaluation protocol, and model interfaces constant across a diverse suite of WHAR datasets. Under these shared assumptions, a clear picture emerges: many contemporary WHAR model architectures have converged toward a similar predictive performance ceiling. Although CNN-HAR attains the highest mean test macro-F1 in the current snapshot, TinyHAR,

TripleCrossDomainAttention, TinierHAR, and several other strong baselines remain very close in aggregate performance. Across 30 datasets, the leading models are separated only by small margins. At the benchmark level, architectural novelty alone therefore no longer appears to translate into substantial predictive gains.

This result is difficult to see in smaller or weakly standardized comparisons. When evaluation is limited to only a few datasets, apparent winners can still emerge, but our per-dataset winner analysis (see [Section 5.2](#)) shows that these wins often do not generalize. The practical implication is that claims of architectural superiority in WHAR should usually be interpreted as claims about a specific setting unless they are supported by large-scale, standardized evaluation. In other words, the state of the art is better characterized by robustness across heterogeneous datasets, sensing setups, and activity spaces than by isolated single-dataset victories.

At the same time, the benchmark is far from saturated. Some datasets are close to ceiling predictive performance, but others remain consistently difficult and still exhibit large spreads between the best and worst models. This suggests that the current plateau reflects unresolved properties of the tasks themselves, including inter-subject variability, ambiguous activity boundaries, inconsistent annotation granularity, sensor-placement differences, and uneven modality coverage. The fact that the leading models plateau around roughly 67 % mean test macro-F1 shows that there is still substantial room for improvement, but that progress is unlikely to come from architecture changes alone. Larger and better-annotated datasets would help cover these sources of variation more completely, but in WHAR such data collection and annotation is expensive and difficult to scale. A more realistic path is to improve how existing models adapt to new subjects and sensing conditions with limited supervision, for example through domain adaptation, personalization, and calibration under distribution shift.

Our benchmark also clarifies which baselines remain credible. Among the classical methods, SVM and k-NN are no longer competitive as general-purpose choices, whereas RandomForest remains surprisingly strong. It is competitive in predictive performance, wins multiple datasets, and occupies the lowest-latency end of the Pareto frontier. This is a useful reminder that stronger WHAR baselines do not always have to be deep architectures. At the same time, several heavier recurrent or hybrid neural model architectures incur substantially higher latency and memory costs without delivering correspondingly better mean test macro-F1. For future comparisons, the most informative baselines are therefore not the most complex ones, but the strongest Pareto-efficient ones: TinierHAR, CNN-HAR, TinyHAR, and RandomForest, depending on the deployment constraints of interest.

Taken together, these findings suggest that the next frontier for WHAR is twofold. Methodologically, the field still needs large-scale standardized benchmark infrastructure to distinguish genuine robustness from improvements that depend on dataset choice or protocol details. Technically, further progress is more likely to come from improvements in efficiency, which, unlike predictive performance, does not yet appear to have reached a comparable ceiling, stronger adaptation to domain shift, and more deployment-aware design than from simply increasing architectural complexity.

6.1 Limitations & Future Work

The current benchmark also has clear boundaries that define how its results should be interpreted. Although the model suite is broad, it is still a curated subset of the WHAR design space rather than an exhaustive collection of all recent architectures. Likewise, the hardware study is anchored to one Android reference device to keep the deployment measurements reproducible, but that choice cannot capture the full range of wearable and edge hardware.

Another important boundary is that the benchmark prioritizes standardized comparison over per-model performance maximization. We evaluate all models, including both deep and classical baselines, under one shared protocol with fixed benchmark-level hyperparameter choices rather than extensive model-specific hyperparameter optimization. The reported results should therefore be read as comparative baseline measurements under shared assumptions, not as a claim that each model has been tuned to its absolute best achievable performance on every dataset.

Future work should extend this foundation in three concrete directions. First, the benchmark should be maintained as a living resource that incorporates newly released datasets and newly competitive baselines, especially lightweight, sequence-based, and multimodal architectures. Second, the hardware study should be expanded to multiple device classes, including smartwatches, microcontrollers, and other low-power edge platforms, with additional attention to energy metrics and architecture-specific deployability constraints. This is important because deployment costs are not architecture-independent constants: different targets expose different memory architectures, execution runtimes, and accelerator capabilities. Our current deployment analysis should therefore be interpreted as a reproducible reference point for one realistic device class, not as a complete characterization of cross-architecture deployability. Third, the open benchmark infrastructure should evolve into a low-friction community workflow in which new datasets and models can be integrated, validated, and compared under the

same transparent assumptions without requiring researchers to rebuild the full pipeline from scratch. Where computationally feasible, future extensions should also study how relative rankings change under stronger model-specific hyperparameter optimization.

7 Conclusion

This paper addresses a central WHAR challenge: the field still lacks a shared and reproducible benchmark that supports fair comparison across datasets, models, and deployment constraints, thereby sustaining the comparability crisis identified earlier in the paper (see Section 2.2). To address this gap, we introduce a unified benchmarking framework comprising three components: the WHAR Datasets library, available at <https://anonymous.4open.science/r/whar-datasets-8141>, which standardizes data formats and processing; the WHAR Models library, available at <https://anonymous.4open.science/r/whar-models/>, which standardizes model architecture interfaces; and a shared cross-subject evaluation protocol. We apply this framework at scale across 30 curated datasets and 17 representative models. By jointly reporting predictive performance and deployment efficiency, the benchmark offers a more realistic reference point for both method development and deployment-oriented model selection.

The empirical results show that the WHAR state of the art is distributed rather than dominated by a single architecture. CNN-HAR achieves the highest mean macro-F1 in the current snapshot, but the strongest models remain tightly clustered and their rankings shift substantially across datasets. Strong benchmark performance in WHAR should therefore be interpreted primarily as robustness across heterogeneous sensing conditions rather than as a win on a single dataset or a marginal advantage in an aggregate table.

The deployment analysis sharpens this picture further. Compact architectures occupy the most compelling performance-efficiency region: TinierHAR is the strongest overall efficiency-oriented neural model, CNN-HAR is the strongest higher-accuracy compact alternative, and RandomForest remains a credible low-latency classical baseline. In contrast, several heavier recurrent or hybrid models incur much larger runtime and memory costs without delivering correspondingly better benchmark-level accuracy. For practical wearable deployment, the benchmark therefore suggests focusing on this small Pareto-efficient subset and validating those candidates under the constraints of the target dataset and device.

More broadly, the main contribution of this work is methodological as much as empirical. The paper answers the gaps identified in the introduction and related work by providing an open, standardized, and reusable benchmark infrastructure together with a large-scale execution under shared assumptions. It does not claim to identify a final winner or to be the last word on WHAR model quality. Instead, WHAR Arena establishes the common experimental ground required to make future progress claims in WHAR more credible, more comparable, and more deployment-relevant.

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A Appendix

A.1 Per-Dataset Latency

The complete per-dataset latency results are reported below as a single matrix with models as columns and datasets as rows.

Dataset	wisdm	14.42	1.62	8.03	16.78	16.18	5.17	4.75	69.79	36.77	1.03	9.33	0.48	6.07	0.80	2.40	2.71	3.24	11.74
	uci_har	47.30	5.39	12.56	99.01	47.18	13.65	6.49	17.50	43.30	1.77	17.91	0.47	9.20	0.71	2.28	9.71	6.26	20.04
	utd_mhad	46.79	4.62	10.87	43.72	40.97	11.41	5.41	2.00	44.07	2.68	8.79	0.73	11.03	0.39	3.00	5.78	9.26	14.80
	hapt	43.58	4.57	9.54	42.97	43.80	11.38	8.56	18.77	55.68	1.60	8.54	0.48	14.10	0.56	2.17	5.09	5.85	16.31
	usc_had	148.21	8.04	13.75	96.73	143.45	21.43	9.53	33.10	97.59	2.22	14.73	0.50	14.17	1.12	3.72	15.99	7.89	37.19
	unimib_shar	46.43	2.28	9.34	34.49	39.78	9.21	2.88	14.61	56.76	1.82	8.46	0.71	9.39	0.65	1.79	11.74	4.69	15.00
	motion_sense	66.13	8.77	21.19	60.58	44.06	23.46	16.81	39.05	56.65	2.03	13.35	0.60	9.82	1.48	2.39	8.26	12.14	22.75
	real_life_har	19.17	6.03	10.91	17.72	21.35	11.32	18.09	738.61	25.80	1.33	8.40	0.53	6.98	9.66	2.39	4.93	6.23	53.50
	wisdm_19_phone	17.84	1.89	7.92	32.15	13.99	6.70	5.18	174.34	24.69	0.99	7.45	0.67	5.23	4.40	2.01	3.71	3.76	18.41
	wisdm_19_watch	19.46	2.09	7.65	13.76	15.46	6.45	4.01	188.29	29.82	1.57	5.95	0.53	4.95	6.15	1.46	2.62	5.25	18.56
	hang_time	35.96	3.37	7.51	42.07	46.44	10.95	4.15	99.53	59.33	3.14	7.17	0.49	13.04	0.79	3.55	4.73	4.62	20.40
	pamap2	173.68	31.75	60.60	129.09	144.88	81.85	51.76	37.85	105.40	10.64	47.42	1.32	26.26	2.79	6.65	19.14	25.38	56.26
	opportunity	104.99	26.27	70.06	77.43	76.16	93.23	207.62	36.01	37.12	47.25	55.07	0.91	13.99	2.36	5.61	13.21	17.22	52.03
	hhar	18.47	2.38	7.62	16.47	15.68	6.34	3.83	94.16	25.63	1.26	6.92	0.46	5.64	1.27	1.55	1.84	4.09	12.57
	mhealth	64.57	9.40	24.47	46.26	63.50	30.05	17.73	9.26	56.39	2.41	15.12	0.62	10.84	0.80	2.80	9.04	13.07	22.14
	dsads	37.29	9.96	25.90	37.48	38.33	26.40	36.25	76.39	29.45	3.48	15.05	0.64	7.65	3.27	5.73	6.21	12.41	21.88
	sad	105.33	16.39	44.97	69.81	91.97	56.24	58.28	24.94	56.64	10.39	24.66	1.10	8.21	1.80	6.35	13.02	17.22	35.73
	daphnet	59.77	8.80	13.69	54.54	60.66	18.46	7.42	23.82	67.36	2.62	10.42	0.48	9.64	0.53	7.02	8.35	7.57	21.24
	real_world	129.72	29.23	58.17	94.56	85.72	72.33	99.70	138.07	50.43	25.29	41.16	1.13	15.25	7.98	6.92	14.21	24.20	52.59
	up_fall	30.65	8.17	23.13	29.66	27.82	20.53	36.58	26.47	29.45	5.06	15.55	0.70	8.77	1.35	2.78	5.41	7.79	16.46
	uma_fall	369.73	49.15	89.49	323.31	269.90	128.48	46.01	19.07	176.59	21.11	65.87	2.16	48.62	3.13	22.09	28.68	64.27	101.63
	real_disp	131.70	34.39	69.46	121.67	107.14	155.69	220.08	91.73	60.88	28.43	58.19	1.35	24.38	17.20	8.61	19.82	41.98	70.16
	hugadb	72.85	13.84	35.53	70.92	76.54	40.45	33.45	42.28	53.17	6.75	23.93	0.91	14.43	2.38	5.22	16.20	16.51	30.90
	harth	45.89	4.18	7.27	51.24	46.06	11.82	7.34	141.80	57.68	1.94	11.11	0.48	9.62	0.99	2.17	5.96	5.74	24.19
	w_har	237.34	13.58	28.25	252.51	242.62	31.92	11.54	10.31	287.39	31.28	21.58	0.77	39.65	0.93	7.89	27.53	12.49	73.98
	wear	44.19	7.23	14.28	46.41	45.00	18.78	13.61	54.50	57.01	1.61	11.88	0.57	9.21	1.62	2.81	6.45	12.02	20.42
	har70	63.07	7.06	10.90	46.64	37.83	16.86	6.94	54.35	63.03	1.57	8.37	0.52	11.13	0.54	2.55	4.95	4.94	20.07
	uca_ear	28.33	3.07	17.85	23.51	22.25	8.48	6.59	33.53	35.47	0.87	6.76	0.56	9.78	0.98	1.95	4.31	5.31	12.33
	gotov	24.55	7.89	21.81	27.12	25.86	15.83	46.00	100.68	29.75	1.89	9.99	0.51	5.67	5.34	1.64	4.10	7.50	19.77
	actrectut_gestures	25.67	6.86	12.01	24.52	27.65	14.02	8.97	1.12	28.95	1.20	8.94	0.39	6.79	0.39	1.57	4.36	5.59	10.53
	Mean	75.77	11.28	25.16	68.10	65.94	32.63	33.52	80.40	61.27	7.51	18.94	0.73	12.98	2.75	4.30	9.60	12.48	30.79
		Attend+Discriminate	CNN-HAR	DANA	DeepConvLSTM	DeepConvLSTM-Attn	DeepConvShallowLSTM	DynamicWHAR	k-NN	Guan-LSTM	MLP-HAR	MLP-Mixer	Random Forest	SA-HAR	SVM	TinierHAR	TinyHAR	Triple-Cross-Attn	Mean
		Model																	

Fig. 10. Per-dataset latency matrix. Rows correspond to datasets, columns correspond to models, and the lowest latency per dataset is highlighted.

A.2 Per-Dataset Memory

The complete per-dataset memory results are reported below as a single matrix with models as columns and datasets as rows.

Dataset	wisdm	8.67	0.75	1.91	2.73	0.92	2.24	0.00	3.01	3.31	1.19	3.24	9.44	3.08	8.19	0.75	1.05	0.23	2.98	
	uci_har	30.73	1.75	3.71	0.00	5.22	4.37	3.70	2.72	3.32	1.47	2.64	5.88	5.31	8.28	0.00	4.86	1.74	5.04	
	utd_mhad	32.84	1.50	1.01	3.88	3.65	1.46	4.47	0.67	0.00	1.95	4.77	9.50	5.32	1.01	0.00	4.71	3.09	4.70	
	hapt	23.39	1.59	4.54	1.84	4.20	3.19	0.00	3.12	1.46	0.00	4.49	13.75	4.92	5.13	1.17	5.62	2.23	4.74	
	usc_had	69.91	2.20	2.23	4.31	4.88	3.30	4.79	3.09	5.77	4.79	0.00	15.12	6.26	10.57	0.00	2.83	66.59	12.15	
	unimib_shar	30.63	0.15	0.00	2.44	3.32	2.55	4.66	1.54	0.00	0.85	0.00	25.22	4.63	4.26	0.46	2.08	2.34	5.01	
	motion_sense	35.03	3.01	8.11	6.78	5.91	7.09	10.21	9.57	3.08	1.18	4.21	4.26	6.87	11.05	1.21	5.25	0.00	7.23	
	real_life_har	13.00	0.93	8.19	5.04	3.75	4.10	9.04	161.47	3.86	1.85	4.12	17.59	2.54	101.70	0.31	1.60	1.41	20.03	
	wisdm_19_phone	11.83	1.30	3.60	2.86	2.59	2.64	4.41	16.39	2.98	0.09	3.69	32.89	2.36	54.45	0.45	1.41	2.32	8.60	
	wisdm_19_watch	10.62	0.93	3.07	2.79	3.50	2.62	5.12	19.26	3.70	0.14	2.28	36.33	0.00	58.41	0.76	0.58	1.40	8.91	
	hang_time	33.27	0.93	3.43	1.14	3.88	2.15	3.62	4.32	2.56	0.00	4.17	20.53	4.12	8.54	0.00	6.81	1.61	5.95	
	pamap2	68.88	14.43	16.70	9.10	8.89	11.21	30.65	22.17	4.96	12.30	7.15	16.10	8.89	24.06	3.06	17.82	20.17	17.44	
	opportunity	28.71	13.03	45.94	18.79	19.06	23.55	69.20	35.25	3.40	40.84	9.06	7.82	3.21	23.42	2.99	8.96	9.81	21.36	
	hhar	10.21	0.54	4.02	3.17	3.70	1.28	4.27	8.17	3.72	2.47	2.70	14.60	1.99	15.25	0.86	2.47	1.81	4.78	
	mhealth	35.59	4.50	9.54	8.30	8.74	7.55	8.89	4.15	4.92	2.71	3.83	5.69	5.25	8.82	1.02	4.70	5.59	7.63	
	dsads	21.77	1.73	16.80	11.42	10.14	12.77	21.45	37.44	3.02	6.62	3.21	11.38	5.34	27.09	1.29	0.00	5.57	11.59	
	sad	40.79	12.71	19.41	13.31	14.41	15.95	31.59	15.54	4.83	11.97	6.94	5.54	7.01	15.15	1.39	3.60	8.70	13.46	
	daphnet	40.54	0.32	6.03	4.27	4.52	6.23	3.96	3.35	2.89	3.16	3.76	4.84	6.82	2.79	0.31	5.93	2.47	6.01	
	real_world	38.16	11.41	29.34	12.84	12.52	12.52	38.44	96.94	3.33	17.66	6.34	16.06	4.73	74.56	1.94	10.42	12.39	23.51	
	up_fall	16.70	3.19	13.79	9.55	8.96	8.60	25.61	10.81	4.22	5.26	2.89	13.83	6.96	14.49	0.12	2.98	0.00	8.70	
	uma_fall	145.39	12.18	14.32	9.27	7.57	11.73	26.09	10.80	5.89	18.48	16.29	16.91	10.26	22.08	4.05	29.88	30.88	23.06	
	real_disp	44.40	14.33	37.46	17.27	15.36	17.56	58.09	83.29	4.60	31.32	12.26	27.39	4.96	191.42	2.96	9.94	33.49	35.65	
	hugadb	55.95	7.81	17.23	5.27	18.94	16.35	23.48	20.21	5.35	5.76	6.12	15.19	5.00	15.76	0.90	6.55	8.98	13.82	
	harth	31.25	0.68	2.36	3.85	3.96	2.57	4.23	13.91	4.88	0.71	4.97	15.21	5.35	9.37	0.83	3.90	2.67	6.51	
	w_har	173.88	4.51	1.47	10.42	8.23	13.53	0.00	2.85	6.01	21.36	6.30	5.85	10.62	3.56	1.42	29.38	6.51	17.99	
	wear	33.24	3.51	5.96	6.31	6.02	5.93	0.00	9.63	3.70	2.39	3.54	23.50	5.36	16.69	0.29	4.79	4.25	7.95	
	har70	27.94	1.04	4.21	3.98	4.83	3.61	3.78	5.19	3.09	1.85	6.02	3.95	5.55	6.15	1.11	5.26	2.31	5.28	
	uca_ear	12.98	0.17	6.28	3.54	3.52	3.41	3.21	3.55	2.64	0.25	4.38	17.90	3.26	10.64	0.61	1.46	1.07	4.64	
	gotov	12.93	1.74	8.92	5.59	6.38	5.39	15.96	37.88	3.43	3.38	3.19	20.11	4.33	56.43	0.38	1.29	2.09	11.14	
	actrectut_gestures	25.56	3.00	10.96	3.58	6.20	6.54	71.84	2.00	4.59	1.93	4.28	5.09	3.30	3.11	1.05	3.93	2.82	9.40	
	Mean		38.83	4.20	10.35	6.45	7.13	7.40	16.36	21.61	3.65	6.80	4.90	14.58	5.12	27.08	1.06	6.34	8.15	11.18
			Attend+Discriminate	CNN-HAR	DANA	DeepConvLSTM	DeepConvLSTM-Attn	DeepConvShallowLSTM	DynamicWHAR	k-NN	Guan-LSTM	MLP-HAR	MLP-Mixer	Random Forest	SA-HAR	SVM	TinierHAR	TinyHAR	Triple-Cross-Attn	Mean
		Model																		

Fig. 11. Per-dataset memory matrix. Rows correspond to datasets, columns correspond to models, and the lowest memory overhead per dataset is highlighted.

A.3 Per-Dataset FLOPs

The complete per-dataset FLOPs estimates are reported below as a single matrix with models as columns and datasets as rows.

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Dataset	wisdm	41.15M	2.19M	27.30M	44.39M	45.82M	31.38M	4.31M	2.24M	96.03M	265.8k	20.80M	3.6k	52.12M	1.29M	239.2k	2.02M	3.16M	22.04M
	uci_har	287.56M	20.27M	176.77M	288.36M	292.78M	375.44M	30.16M	1.62M	241.92M	5.22M	104.37M	22.3k	147.33M	1.05M	1.48M	16.01M	33.18M	119.03M
	utd_mhad	204.57M	13.61M	118.92M	210.22M	214.65M	255.92M	15.09M	113.4k	242.62M	3.11M	103.76M	14.6k	145.86M	113.5k	1.04M	11.23M	22.49M	91.96M
	hapt	204.57M	13.60M	118.92M	210.22M	214.64M	255.50M	15.08M	1.20M	241.46M	3.11M	103.76M	14.6k	145.84M	676.5k	1.04M	11.23M	22.26M	91.95M
	usc_had	430.14M	28.88M	231.47M	442.11M	451.53M	581.25M	17.68M	2.30M	482.93M	11.24M	204.44M	27.7k	338.46M	1.61M	2.09M	23.78M	44.47M	193.79M
	unimib_shar	122.20M	6.92M	61.07M	132.07M	136.50M	135.52M	5.09M	479.7k	240.93M	1.37M	52.08M	7.3k	144.34M	479.0k	592.1k	6.53M	11.25M	62.20M
	motion_sense	540.27M	40.32M	350.31M	522.79M	527.21M	735.79M	107.50M	7.18M	244.69M	13.80M	259.42M	46.5k	151.83M	3.16M	2.81M	30.78M	66.20M	212.01M
	real_life_har	154.66M	10.68M	123.64M	151.12M	152.56M	145.91M	72.31M	123.00M	97.44M	2.78M	82.25M	16.9k	54.52M	26.22M	956.2k	8.19M	15.56M	71.87M
	wisdm_19_phone	69.22M	4.32M	51.39M	71.08M	72.51M	60.08M	13.53M	11.69M	96.77M	678.4k	41.21M	6.7k	52.74M	11.23M	419.3k	3.54M	6.30M	33.33M
	wisdm_19_watch	69.22M	4.32M	51.39M	71.08M	72.51M	60.08M	13.53M	13.67M	96.77M	678.4k	41.21M	6.7k	52.74M	11.80M	419.3k	3.54M	6.30M	33.49M
	hang_time	122.19M	6.91M	61.07M	132.07M	136.50M	135.21M	5.09M	3.40M	240.08M	1.37M	52.08M	7.3k	144.33M	1.87M	591.4k	6.53M	11.16M	62.38M
	pamap2	3.25G	246.47M	1.98G	2.96G	2.96G	4.76G	925.95M	14.48M	511.34M	199.41M	1.33G	283.0k	384.47M	5.24M	15.84M	196.15M	383.27M	1.18G
	opportunity	2.38G	162.60M	1.57G	1.97G	1.98G	2.63G	7.17G	25.51M	167.96M	244.33M	1.04G	244.3k	119.99M	5.50M	11.83M	152.64M	232.23M	1.17G
	hhar	69.22M	4.31M	51.39M	71.08M	72.51M	60.01M	13.53M	5.89M	96.40M	676.7k	41.20M	6.7k	52.72M	3.71M	418.5k	3.53M	6.26M	32.52M
	mhealth	683.08M	51.46M	446.72M	653.03M	657.45M	936.18M	172.52M	2.22M	246.76M	20.02M	311.51M	60.5k	154.34M	1.10M	3.55M	39.29M	84.95M	262.60M
	dsads	599.29M	43.06M	448.88M	552.65M	554.58M	663.14M	657.80M	28.53M	126.99M	25.67M	285.87M	64.9k	76.84M	7.38M	3.33M	34.58M	69.98M	245.80M
	sad	1.79G	133.86M	1.16G	1.62G	1.62G	2.42G	1.22G	10.88M	257.67M	98.25M	778.68M	168.9k	172.84M	3.28M	9.03M	108.66M	220.45M	683.95M
	daphnet	376.25M	26.58M	224.04M	377.30M	383.13M	509.16M	31.24M	2.21M	309.36M	7.86M	131.76M	28.1k	196.97M	356.6k	1.91M	20.91M	40.49M	155.27M
	real_world	2.47G	180.63M	1.57G	2.16G	2.17G	3.26G	2.35G	74.74M	264.19M	167.91M	1.04G	233.0k	183.34M	18.66M	12.14M	153.10M	297.76M	962.30M
	up_fall	422.01M	29.81M	340.38M	391.28M	392.71M	403.66M	567.68M	8.20M	100.98M	16.93M	225.57M	49.5k	59.93M	3.35M	2.57M	23.30M	43.61M	178.35M
	uma_fall	4.95G	381.59M	2.95G	4.60G	4.62G	7.59G	547.23M	5.67M	1.01G	378.89M	2.05G	409.8k	930.13M	4.61M	23.90M	296.16M	599.01M	1.82G
	real_disp	3.69G	260.82M	2.26G	3.10G	3.11G	4.70G	5.35G	63.47M	277.26M	329.95M	1.51G	345.5k	201.38M	48.11M	17.47M	237.82M	437.82M	1.51G
	hugadb	1.37G	103.61M	878.53M	1.27G	1.28G	1.93G	474.22M	14.66M	301.55M	58.72M	614.47M	123.2k	199.82M	4.10M	6.99M	80.31M	167.37M	514.82M
	harth	204.57M	13.60M	118.92M	210.22M	214.64M	255.50M	15.08M	9.81M	241.46M	3.11M	103.76M	14.6k	145.84M	1.90M	1.04M	11.23M	22.26M	92.53M
	w_har	1.26G	87.29M	663.44M	1.28G	1.30G	1.80G	31.63M	853.5k	1.21G	75.72M	512.86M	79.4k	1.20G	402.3k	5.96M	70.52M	136.22M	566.48M
	wear	371.17M	26.96M	234.62M	366.50M	370.93M	495.90M	50.48M	6.99M	243.77M	7.71M	156.05M	30.2k	148.85M	3.64M	1.92M	20.86M	44.55M	150.05M
	har70	204.56M	13.59M	118.92M	210.21M	214.64M	255.35M	15.08M	3.44M	241.08M	3.11M	103.76M	14.6k	145.83M	790.6k	1.04M	11.23M	22.18M	92.05M
	uca_ehar	104.21M	6.78M	72.55M	106.02M	107.95M	107.25M	18.02M	2.53M	120.89M	1.23M	52.18M	9.4k	67.33M	2.05M	579.4k	5.58M	10.89M	46.24M
	gotov	281.32M	19.90M	228.00M	266.76M	268.19M	270.06M	246.99M	27.99M	99.41M	8.13M	143.80M	32.3k	57.13M	13.69M	1.73M	15.25M	29.16M	116.33M
	actrectut_gestures	274.96M	19.80M	191.17M	268.53M	271.16M	333.91M	73.87M	168.9k	156.31M	5.13M	131.37M	25.2k	90.83M	138.2k	1.53M	15.14M	33.95M	109.88M
	Mean	899.77M	65.49M	562.52M	823.56M	828.57M	1.20G	674.50M	15.84M	276.67M	56.55M	387.29M	79.9k	200.61M	6.25M	4.48M	53.99M	104.16M	362.65M
		Attend+Discriminate	CNN-HAR	DANA	DeepConvLSTM	DeepConvLSTM-Attn	DeepConvShallowLSTM	DynamicWHAR	k-NN	Guan-LSTM	MLP-HAR	MLP-Mixer	Random Forest	SA-HAR	SVM	TinierHAR	TinyHAR	Triple-Cross-Attn	Mean
		Model																	

Fig. 12. Per-dataset FLOPs matrix with datasets as rows and models as columns. The lowest forward-pass compute per dataset is highlighted.